

# APT-BENCH: PATIENT-DISJOINT CELL TYPING FROM SINGLE-CELL APTAMER PROFILES

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## ABSTRACT

Single-cell aptamer profiling offers a blood-derived molecular input for immune-cell identification, but evaluating it requires separating performance on additional cells from generalization to new patients. We introduce APT-Bench, a benchmark of 361,792 cells with 293 aptamer measurements from 40 participants across five cancer groups and healthy controls. Paired transcriptomic annotations define five lineages and 27 subtypes. Five fixed patient-disjoint folds and subject-balanced evaluation compare linear, tree-based, and Transformer references on the same APT-only tasks. We describe a Soft-cascade Transformer reference that combines global subtype prediction with softly routed lineage experts, without assuming a model-specific advantage. The benchmark exposes substantial remaining difficulty in fine typing; a separate matched-input RNA diagnostic contextualizes, rather than replaces, the APT comparison. As a downstream case study, nested cell-output aggregation recovers patient subtype composition better than a training-mean reference, while incremental disease-prediction value remains unresolved. APT-Bench establishes a framework for comparing models of this measurement modality and studying their patient-level use, with clinical interpretation requiring acquisition-aware external validation.

## 1 INTRODUCTION

A blood sample contains information about its constituent immune cells and variation associated with the person from whom it was collected. A single-cell assay can therefore be asked two connected questions: which cells are present, and what do their measurements reveal about the patient’s disease state? Single-cell aptamer profiling offers a setting in which to evaluate both questions using the same molecular measurements. The long-term motivation is minimally invasive blood-based characterization; reliable diagnosis from a specified small blood volume is an eventual validation target, not an outcome established by this study.

The primary aim of this paper is a benchmark for APT-only cell typing in new patients, with a documented hierarchy-aware reference model. Patient-level disease characterization is a secondary application, not an equally validated clinical endpoint. Moving from the motivation to an evaluation requires separating cell identity from patient state. A model can distinguish broad immune lineages without reconstructing transcriptome-defined subtypes. Conversely, a disease classifier can exploit patient-level measurement offsets without learning cell composition. High performance at one level therefore need not validate the other. Correlated cells also cannot substitute for independent patients when estimating disease generalization. These distinctions motivate a benchmark that connects the two levels while testing the limits of their interpretation.

We introduce APT-Bench, comprising 361,792 labeled single-cell profiles from 40 participants in five cancer groups and healthy controls. Each cell has 293 aptamer measurements and paired-transcriptome-derived labels in a five-lineage, 27-subtype ontology. The benchmark evaluates APT-only cell typing and within-cohort patient characterization under fixed patient-disjoint folds. A Soft-cascade Transformer combines global subtype prediction with softly routed lineage experts as one reference alongside linear, tree-based, and other Transformer models. Benchmark validity does not depend on that model winning, and the current comparisons do not isolate its component effects. A separate paired-RNA diagnostic asks how reconstructability depends on the input modality, while

strictly nested cross-fitting connects predicted cell profiles to patient-level disease labels without exposing test patients during either learning stage (Figure 1).

The results establish reference performance for a targeted molecular input rather than a diagnostic success claim. Fine typing remains difficult under the evaluated models. A matched-input probe reconstructs the RNA-derived labels more strongly from RNA; that modality contrast is not evidence of algorithm superiority on identical inputs. In the secondary application, predicted subtype summaries improve reference-composition recovery over a training-mean baseline. Disease association remains a distinct question, with unresolved incremental subtype value and acquisition-related explanations.

Reliability is therefore part of the biological question, not an independent promise of clinical validity. Patient-disjoint splits, subject-balanced metrics, and acquisition-sensitivity controls delimit what can be concluded from this cohort. Supporting sampling experiments on APT, COMBAT, and OneK1K further show why subject count, cell count, and training label distribution must be distinguished. These external datasets evaluate sampling behavior; they do not independently validate APT disease prediction.

Our contributions are:

- **An APT cell-typing benchmark.** We specify a paired-annotation dataset, fixed patient partitions, ontology, and patient-aware evaluation for a 293-channel single-cell aptamer input.
- **A reference-model suite.** Classical and Transformer recipes, including a documented Soft-cascade reference, establish APT-only comparison points; a separate paired-input diagnostic assesses modality dependence.
- **A cell-to-patient application study.** Nested aggregation tests reference-composition recovery and exploratory disease association, with explicit controls for alternative sources of predictive signal.

## 2 RELATED WORK

**Blood-based characterization.** Multi-cancer detection commonly uses circulating DNA, methylation, fragmentomics, or bulk proteins (Cohen et al., 2018; Zeng et al., 2026; Wan et al., 2025; Kahwati et al., 2025). Our question is complementary: whether single-cell aptamer measurements can support both immune-cell identification and patient disease-associated characterization. APT-Bench evaluates that connection within a research cohort; it does not equate within-cohort classification with prospective screening.

**Single-cell representations and hierarchy.** Geneformer, scGPT, and scFoundation motivate large transcriptomic representations, while controlled studies show that simple baselines can remain competitive (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024; Kedzierska et al., 2025). Cell annotation also motivates hierarchical losses and architectures (Jia et al., 2025; Cultrera di Montesano et al., 2026). We use flat, HCE, and soft-cascade Transformers alongside classical models as APT references. Because their complete objectives are not matched, this ladder illustrates the evaluation setting rather than constituting a new learning algorithm or component ablation.

**Evaluation across cells and subjects.** Reusable single-cell benchmarks separate datasets, models, and metrics and adopt sample- or study-disjoint evaluation (Luecken et al., 2025; Liu et al., 2025; Shitov et al., 2025; Liu et al., 2026). APT-Bench links cell and patient tasks with nested cross-fitting and makes their statistical units explicit. Gradient-boosted trees provide strong tabular references (Grinsztajn et al., 2022; Erickson et al., 2026). Supporting sampling and feature-budget audits address correlated observations and resource choices, but cannot resolve biological versus technical confounding.

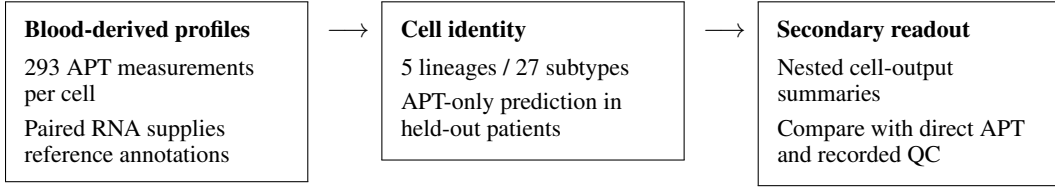


Figure 1: From cell identity to patient characterization. The cell-to-patient route uses nested cross-fitting; a parallel direct-APT route summarizes measurements without requiring cell labels. Patient-disjoint splits prevent identity overlap, while centering and QC controls assess alternative sources of disease association. Neither route constitutes validated clinical diagnosis.

### 3 BENCHMARK SETTING

#### 3.1 BLOOD PROFILES AND REFERENCE LABELS

APT-Bench is derived from peripheral-blood specimens from 40 participants in five cancer groups (CRC, GC, HCC, PC, and BTC) and healthy controls, with 6–8 participants per group. The paired assay measures transcriptomic state and 293 target-specific nucleic-acid aptamers on the same cells. Transcriptomic counts were processed with Seurat (Hao et al., 2021); predicted doublets and low-quality cells were removed, and clusters were manually annotated using canonical markers from prior PBMC and pan-cancer studies (McGinnis et al., 2019; Zhang et al., 2024; Yang et al., 2024). These annotations define five lineages and 27 subtypes. They are operational reference labels, not independently measured biological truth.

The linked post-QC cohort contains 374,854 cells. Excluding the uncertain *Unknown* subtype leaves 361,792 benchmark cells. Each subtype has one fixed parent lineage. Disease is a six-class patient attribute, not the root of the cell hierarchy: any given lineage can occur in multiple disease groups. The benchmark therefore connects cell and patient tasks through aggregation, rather than imposing a disease-to-lineage-to-subtype label tree.

The construction checklist and unresolved acquisition and annotation fields are recorded in Appendix C.1. In particular, the analyzed data do not establish a validated one-drop input-volume requirement. The scope here is characterization from the profiled blood cells, not end-to-end validation of a small-volume diagnostic assay.

#### 3.2 FROM CELL IDENTITY TO PATIENT STATE

The evaluation has a primary task and a secondary application. **Cell identity** predicts lineage and subtype from APT alone. **Patient characterization** predicts disease-group membership from patient summaries of measurements or cell-model outputs. The latter is an exploratory association task: with one acquisition per participant and incomplete acquisition metadata, disease biology and technical structure cannot be identified separately. Open-set subtype discovery is outside the contract; all primary cell tasks use the fixed five-lineage/27-subtype ontology.

Five frozen disease-stratified folds each hold out eight patients. All cells from a patient remain together; preprocessing, model selection, and fitting are restricted to development patients. We pool predictions across the five out-of-fold (OOF) partitions. A cell-random split is retained only as a leakage diagnostic, not as evidence of generalization to new patients. The release interface records cell and patient IDs, ontology, folds, predictions, and metrics (Appendix C.6). Full data access and a portable end-to-end evaluator remain release requirements.

#### 3.3 METRICS AND STATISTICAL UNITS

Coarse and fine subject-balanced pooled Macro-F1 (SB-Macro-F1) are the primary cell metrics. For patient  $p$ , let  $C^{(p)}$  be the confusion matrix over a fixed  $K$ -class ontology and  $n_p$  its number of

evaluated cells. Define

$$\bar{C} = \frac{1}{P} \sum_{p=1}^P \frac{C^{(p)}}{n_p}, \quad \text{SB-MacroF1} = \frac{1}{K} \sum_{k=1}^K \frac{2\bar{C}_{kk}}{\sum_j \bar{C}_{kj} + \sum_i \bar{C}_{ik}}. \quad (1)$$

A zero denominator contributes zero. Each patient’s entire matrix, not each class row, is normalized to unit mass. This differs from cell-weighted pooled Macro-F1 and from mean within-subject Macro-F1. We retrospectively re-score the frozen OOF predictions under this primary reporting convention; checkpoints are not reselected. Historical cell-weighted scores remain in Table 26.

Main-table intervals use 2,000 paired patient-bootstrap resamples, with the same patient multiplicities across models. They are pointwise, conditional on the fitted models and fixed fold design, and do not include model-selection or training-seed uncertainty. The main comparison is descriptive; an interval for an individual score is not a test of pairwise superiority. Historical cell-weighted tests against HCE and XGBoost remain separately documented and are not transferred to the new metric. Exact-path accuracy, subtype-tree distance, root-excluded hierarchical F1, and consistency are appendix diagnostics.

Patient characterization uses one prediction per patient, with patient-level accuracy, macro-F1, and macro-AUROC. Its effective disease sample is 40, not 361,792. Nested model selection prevents direct test exposure but does not remove confounding or uncertainty from exploratory analysis choices.

The paired-RNA probe also uses subject-balanced pooled Macro-F1, but has a different training budget and preprocessing protocol; its scores must not be interpreted as additional architecture rows in the main table. Appendix C.10 gives the definitions.

### 3.4 SUPPORTING RELIABILITY ANALYSES

The paired-RNA analysis is an internal reconstruction diagnostic, since RNA also supplies the reference annotations. In the current matched-input probe, HVG selection is fitted separately within inner and outer training subsets; this prevents feature-selection exposure to their evaluation patients, but does not make RNA an independent validation of RNA-derived labels. Centering, patient-identity prediction, and recorded QC controls probe alternative explanations for disease association. Feature-panel and inference-cell subsampling are secondary redundancy diagnostics, not evidence of reduced training requirements or clinically efficient assays.

Subject-cell sampling controls are reported separately in Appendix C.11. They compare natural fixed-total allocation with class-quota-matched allocation and distinguish stability across training subsets from confidence intervals for mean effects. COMBAT CITE-seq and OneK1K provide external sampling evaluations (COVID-19 Multi-omics Blood Atlas (COMBAT) Consortium, 2022; Yazar et al., 2022), not external disease validation of the APT assay. Their RNA experiments use restricted 293-gene representations.

## 4 REFERENCE MODELS AND PATIENT READOUTS

### 4.1 SOFT-CASCADE TRANSFORMER

The first APT-Bench contract is APT-only prediction of immune-cell identity in held-out patients. We therefore include a hierarchy-aware neural reference as a documented reference model in the main text. Each cell is represented by a 293-dimensional standardized APT vector. The Transformer backbone embeds aptamer identity and value as tokens, prepends a learnable [CLS] token, and uses the final [CLS] state  $z$  as the cell representation.

Let  $M = 5$  lineages and  $K = 27$  subtypes define the fixed ontology, with parent map  $m(k)$ . The soft-cascade reference predicts lineage logits  $c = W_c z$  and routing probabilities  $p(m | x) = \text{softmax}(c)_m$ . Subtype logits combine a global subtype path and lineage-specific experts:

$$s_k(x) = g_k(z) + \alpha p(m(k) | x) e_{m(k),k}(z) + \gamma \log\{\epsilon + (1 - \epsilon)p(m(k) | x)\}. \quad (2)$$

We use  $\alpha = 1$ ,  $\epsilon = 0.1$ , and ramp  $\gamma$  from 0 to 0.5 during early training. The loss is

$$\mathcal{L} = \mathcal{L}_{\text{fine}} + 0.5 \mathcal{L}_{\text{coarse}} + w(t) \mathcal{L}_{\text{contrast}}, \quad (3)$$

where  $w(t)$  ramps to 0.2. The contrastive term uses same-subtype, cross-disease cells as positives and different-subtype cells as negatives, with additional emphasis on within-lineage negatives. This construction is intended to provide a reproducible hierarchy-aware reference for the benchmark; disease recognition itself is evaluated separately at the patient level.

## 4.2 APT-ONLY BASELINE COMPARISONS

The neural ladder compares the soft cascade with a flat Transformer and the same Transformer trained with a two-level HCE adaptation (Cultrera di Montesano et al., 2026). These references share the APT tokenizer, encoder dimensions, optimizer, training budget, patient folds, training-only preprocessing, and validation fine-macro-F1 checkpoint rule. Their objectives are not fully matched: only the soft-cascade model uses disease labels for cross-disease contrastive pairing, whereas Flat and HCE use classification losses alone. Thus the ladder is a benchmark reference comparison, not a component ablation; no routing, HCE, or contrastive effect is claimed in isolation. Appendix B records implementation details and the component matrix.

Baseline coverage is intentionally distinguished from input compatibility. LR and XGBoost supply linear and nonlinear tabular comparisons in the main APT-only table, alongside the three Transformer recipes. Linear SVM, Random Forest, and reference correlation are retained in the complete appendix inventory. These are not an exhaustive model leaderboard. Plain MLP/ResNet and additional modern tabular learners are applicable to the APT matrix but lack completed full hierarchy evaluations in this release; we do not classify their absence as incompatibility or underperformance. HCE supplies a hierarchy-loss reference, not comprehensive coverage of single-cell annotation systems.

Pretrained transcriptomic encoders such as scGPT and Geneformer use gene-based inputs (Cui et al., 2024; Theodoris et al., 2023). The anonymous APT feature IDs have no verified public gene-token mapping, so direct use of those checkpoints is not defined for the APT-only task. Learning an APT adapter would introduce an additional training protocol, while supplying paired RNA would change the input modality. On the external RNA cohorts, transcriptomic encoders are applicable but unevaluated here, rather than intrinsically incompatible. We make no claim that computational expense alone excludes them, nor that unrun models underperform the reported references.

## 4.3 SECONDARY PATIENT-LEVEL READOUTS

**Nested cell-to-patient bridge.** For the composition bridge, each disease outer fold seals eight patients. A final disease-label-free XGBoost subtype model trained on the other 32 patients constructs outer-test compositions. Downstream training compositions for those 32 patients are instead produced by four inner patient-disjoint fits, each trained on 24 patients. A class-weighted multinomial LR, with regularization selected by patient-level inner CV, is then evaluated once on the sealed eight patients. Thus neither the cell-typing stage nor the disease stage uses an outer-test patient during preprocessing, feature construction, fitting, or selection. Appendix B.4 provides the complete data flow and controls. A conditional increment audit uses the same nested features and tuning protocol to compare recorded QC plus predicted lineage against recorded QC plus predicted subtype composition; subtype probabilities implicitly contain their lineage sums.

**Direct measurement summaries and controls.** Patient-median APT profiles provide a route to disease-group prediction that does not require inferred cell identities. Cell-level disease-source predictions with patient majority voting form a separate diagnostic. We compare these associations with patient-wise centering, patient-identity recognition, and recorded QC summaries rather than treating high within-cohort scores as clinical validation. Full settings are in Appendix C.7.

# 5 RESULTS

## 5.1 APT-ONLY CELL TYPING IN HELD-OUT PATIENTS

Table 1 compares the frozen APT-only reference predictions using subject-balanced pooled Macro-F1. The table addresses model performance on the same measurement modality and label ontology, not whether APT outperforms RNA. All evaluations include the same 40 held-out patients

| Model                    | Input | Coarse SB-Macro-F1 $\uparrow$ | Fine SB-Macro-F1 $\uparrow$ |
|--------------------------|-------|-------------------------------|-----------------------------|
| Logistic Regression      | APT   | 0.298 [0.281, 0.315]          | 0.126 [0.107, 0.139]        |
| XGBoost                  | APT   | 0.325 [0.302, 0.345]          | 0.137 [0.120, 0.149]        |
| Flat Transformer         | APT   | 0.326 [0.300, 0.348]          | 0.139 [0.114, 0.155]        |
| HCE Transformer          | APT   | 0.335 [0.310, 0.356]          | 0.143 [0.121, 0.160]        |
| Soft-cascade Transformer | APT   | 0.332 [0.306, 0.354]          | 0.145 [0.122, 0.160]        |

Table 1: Primary APT-only cell-typing comparison: five patient-disjoint folds, 40 patients, 361,792 cells. Entries are subject-balanced pooled Macro-F1 with pointwise 95% patient-bootstrap intervals (2,000 resamples). Frozen predictions are retrospectively re-scored with equal total patient weight; checkpoints are unchanged. Comparisons are descriptive, conditional on fitted models. Transformer auxiliary supervision differs (Table 3); full-budget MLP coverage remains unavailable.

| Probe      | Input     | Coarse SB-Macro-F1 | Fine SB-Macro-F1 |
|------------|-----------|--------------------|------------------|
| Linear SGD | APT       | 0.277              | 0.090            |
| Linear SGD | RNA       | 0.742              | 0.431            |
| Linear SGD | APT + RNA | 0.751              | 0.458            |
| MLP        | APT       | 0.310              | 0.094            |
| MLP        | RNA       | 0.871              | 0.584            |
| MLP        | APT + RNA | 0.873              | 0.606            |

Table 2: Separate matched-input modality reference, not an APT-only model leaderboard. Inputs share paired cells, patient folds, and a capped training budget; RNA feature selection and standardization are training-subset-only. RNA also underlies the reference labels, so these are reconstruction results, not independent annotation validation. Full pointwise patient intervals and noise/shuffling controls are in Table 4.

across the five folds. Coarse/fine SB-Macro-F1 is 0.298/0.126 for LR and 0.325/0.137 for XGBoost. Flat, HCE, and Soft-cascade Transformers reach 0.326/0.139, 0.335/0.143, and 0.332/0.145, respectively. The cascade has the highest fine point estimate among these rows, while HCE has the highest coarse estimate; we do not infer a model-specific advantage from these rankings. Pointwise patient-bootstrap intervals describe uncertainty in each fitted recipe’s performance; they are not evidence that the cascade’s individual components improve prediction.

The full reference inventory and historical cell-weighted results are in Tables 25 and 26. A plain MLP has been evaluated in the separate matched-input probe below, but not under the full-training hierarchy protocol; it is not substituted into this table. This missing full-budget reference is a coverage limitation, not evidence of incompatibility or underperformance.

All reference models have lower raw fine than coarse F1. The five- and 27-class tasks differ in class count, imbalance, and annotation boundaries, so this difference is descriptive rather than an assay-resolution limit. Class-level support and hierarchy errors are reported in Appendix C.9. The contribution is a common evaluation setting and reference suite, not a required cascade victory.

## 5.2 A SEPARATE PAIRED-INPUT MODALITY REFERENCE

To distinguish input modality from architecture, we compare APT, paired RNA, and their concatenation with the same two probe families, paired cells, patient folds, and training budget (Table 2). The probe uses up to 1,000 training cells per development patient and selects RNA HVGs separately inside each inner and outer training subset. Although the metric matches Table 1, preprocessing, training budget, and model recipes differ; this is not an extension of the APT-only leaderboard.

RNA reconstructs the operational labels more strongly than APT for both probes. For the MLP, adding APT to RNA improves fine Macro-F1 by +0.022 (95% paired patient-bootstrap CI [0.013, 0.037]); for the linear SGD probe the increment is +0.028 ([0.013, 0.041]). The MLP combined input also exceeds RNA plus within-patient shuffled APT by +0.010 ([0.002, 0.017]), whereas this contrast is unresolved for SGD. Full noise and pairing controls are in Appendix C.2.

These results support complementary association under a matched probe, not APT superiority over RNA or an information-theoretic limit of the assay. Training-fold feature selection addresses pre-processing exposure, but RNA also supplies the reference annotations and therefore cannot independently validate their biological accuracy. Chance-adjusted resolution diagnostics are retained in Appendix C.3.

### 5.3 SECONDARY APPLICATION: PATIENT-LEVEL SUMMARIES

We first evaluate patient composition without using disease labels. Nested soft subtype summaries have equal-patient, equal-subtype mean absolute error 0.0148, versus 0.0231 for the outer-training patients’ mean reference composition. The paired error reduction is 0.0084 (95% patient-bootstrap CI [0.0065, 0.0101]). This establishes improvement over a training-only constant reference for composition reconstruction, not a disease-prediction benefit. The lineage reduction is unresolved. Definitions and coverage diagnostics are in Appendix C.8.

As an exploratory disease readout, nested predicted subtype composition reaches patient Macro-F1 0.593 versus 0.477 for predicted lineage composition. Their paired difference is unresolved (+0.116;  $[-0.015, 0.262]$ ), and conditioning both representations on recorded QC reduces the increment to +0.016 ( $[-0.120, 0.169]$ ). Table 9 reports reference compositions and nuisance baselines. Thus cell-model outputs carry disease-associated information, but do not establish incremental fine-subtype utility or a relationship between improved typing and improved diagnosis.

Direct APT summaries show stronger within-cohort association: patient-median APT plus LR reaches accuracy 0.950. However, centering reduces a separate cell-level LR majority-vote readout from 40/40 to 26/40 correct patients, and an identity diagnostic reaches 88.4% on held-out cells from known patients. These controls implicate patient-specific expression structure without identifying its biological or technical origin. The independent disease sample remains 40; acquisition metadata and external APT validation are outstanding. Full controls and secondary budget diagnostics are in Appendices C.7 and C.21.

### 5.4 SUPPORTING EVALUATION AUDITS

Cell-random splitting inflates historical cell-weighted coarse/fine Macro-F1 by +0.090/ + 0.108 for the soft cascade and +0.050/ + 0.069 for XGBoost. This motivates patient-disjoint evaluation independently of the choice to balance patients at scoring time. Equal-cell subsampling, error-conditioned hierarchy nulls, and class-level diagnostics remain available in the appendix.

Sampling studies on APT, COMBAT, and OneK1K examine how subject breadth, cell depth, and training-label allocation change performance. Natural fixed-total and class-quota-matched policies estimate different allocation contrasts; the latter also changes within-class donor allocation. Full grids, uncertainty definitions, extended frontiers, and simulations are in Appendix C.11. They qualify resource-allocation claims and do not constitute external validation of APT disease prediction.

## 6 DISCUSSION

APT-Bench makes single-cell aptamer profiling a defined cell-typing evaluation problem: the input is APT alone, the targets are paired-transcriptomic annotations, and generalization is measured on held-out patients. The reference suite establishes comparison points across linear, tree-based, and Transformer recipes. Soft-cascade Transformer supplies an explicit hierarchy-aware design, but the benchmark does not depend on demonstrating its superiority or presenting its components as novel algorithms.

**Interpreting model and modality comparisons.** The main table holds the measurement modality fixed; the matched-input probe instead holds the probe protocol fixed while varying APT and RNA. These questions must not be conflated. RNA-derived annotations favor interpretation of RNA performance as internal label reconstruction, not independent truth validation. Lower fine F1 also reflects different class counts, prevalence, and boundaries. Full-budget MLP/ResNet coverage and objective-matched cascade ablations remain incomplete. The retrospective shift to subject-balanced

reporting changes the estimand without changing model checkpoints; historical cell-weighted scores are preserved for traceability.

**From cell predictions to patient use.** Subtype summaries improve reference-composition recovery over a training-mean baseline, providing a concrete aggregate-information result even when fine cell typing remains difficult. This does not imply an incremental benefit for disease discrimination. Forty participants across six groups provide limited independent evidence, and patient-disjoint folds cannot remove disease-correlated acquisition effects. Centering can remove biological as well as technical variation. A compact panel that preserves this mixed signal is consequently not a validated diagnostic panel, and inference-cell subsampling does not establish reduced training requirements.

**Scope and next validation.** The benchmark is conditional on one APT cohort and its operational taxonomy. Independent annotation review, completed dataset provenance and access documentation, and portable reproduction remain necessary release work. A subsequent clinical study must freeze the assay and model, document input volume and cell recovery, record acquisition batches, and test an independent APT cohort. External RNA datasets broaden the sampling audit, not the clinical evidence for this assay. These boundaries retain blood-based characterization as the motivation without treating one-drop diagnosis as an established result.

## 7 CONCLUSION

APT-Bench establishes patient-disjoint evaluation of lineage and subtype prediction from single-cell aptamer profiles, with classical and neural references including Soft-cascade Transformer. Separate APT-only and paired-input comparisons distinguish architecture performance from modality effects. Nested patient aggregation demonstrates reference-composition recovery while leaving incremental disease utility unresolved. The benchmark provides a basis for improving APT cell models and evaluating their downstream use; acquisition-aware independent validation is required before clinical claims.

### AI USE STATEMENT

Generative-AI assistance through ChatGPT and Codex was used in the revision workflow for research-design suggestions, literature search, manuscript editing, analysis-code development, debugging, experiment orchestration, and scientific figure preparation. AI-generated suggestions were not themselves experimental evidence; reported computational results are tied to saved run artifacts. **TODO: Authors: confirm the complete tool/model inventory and versions, describe the human checks actually performed on code, citations, figures, and statistical claims, and affirm responsibility for the final submission. Automated checks do not substitute for this author verification.**

### ETHICS STATEMENT

APT-Bench is intended for research evaluation, not clinical decision-making. The small single-cohort sample and unresolved acquisition effects preclude claims of clinical validity or demographic fairness. Strong patient-specific signals also create a privacy concern: removing direct identifiers alone does not establish that cell profiles cannot be linked to their source individuals. Any data release must respect participant consent and applicable access terms. **TODO: Data owners: supply the actual ethics approval or exemption and consent terms; confirm de-identification procedures, permitted sharing/access controls, and conflicts of interest. These facts have not been verified from the available analysis artifacts; preserve reviewer anonymity when providing documentation.**

### REPRODUCIBILITY STATEMENT

The repository records the fixed five-fold partition in `benchmark/splits/patient_folds.json`, selected artifact checksums and structural audits in `benchmark/release_audit/`, and artifact-level reproduction commands in `REPRODUCIBILITY.md`. The latter distinguishes rebuilding figures/PDFs from rerunning models. Per-experiment protocol files record sampling settings and seeds; the controlled simulation includes per-replicate results and source hashes. A snapshot of the local analysis environment is provided, but is not a lockfile for all historical training runs.

TODO: Finalize anonymous reviewer access, data permissions and licenses, archival versioning, original training environments, and a portable raw-data-to-results evaluator. The present repository is not yet a complete independently reproducible benchmark release.

#### AUTHOR CONTRIBUTIONS

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

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TODO: Add funding sources and grant numbers, computational resources, data or clinical contributors, and non-author assistance; verify whether any acknowledgment must be omitted or anonymized during double-blind review.

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## A APPENDIX

### B TRANSFORMER REFERENCE IMPLEMENTATION DETAILS

This appendix gives the implementation details for the hierarchy-aware reference ladder introduced in Section 4: a flat Transformer, its HCE counterpart, and Soft-cascade Transformer. The cascade focuses on the cell-level hierarchy, lineage  $\rightarrow$  subtype, while disease-association audits use separate tabular baselines and patient-level aggregation. The released code retains the historical identifier `DropCascade` for artifact compatibility. That identifier denotes a reproducible benchmark baseline, not a claim that its global path, auxiliary coarse objective, soft routing, or contrastive construction independently improves performance.

#### B.1 CELL ENCODER

Each cell is represented by a standardized APT vector  $x \in \mathbb{R}^{293}$ . We embed feature identity and feature value separately. For aptamer index  $i$ , we form a token

$$t_i = \text{Emb}(i) + \text{MLP}_{\text{val}}(x_i), \quad i = 1, \dots, 293. \quad (4)$$

A learnable `[CLS]` token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final `[CLS]` representation

$$z = \text{Encode}(x) \in \mathbb{R}^{256} \quad (5)$$

serves as the shared cell representation for downstream tasks.

#### B.2 SOFT-CASCADE REFERENCE

Let  $M = 5$  be the number of coarse lineages and  $K = 27$  the number of fine subtypes. Each subtype  $k$  belongs to exactly one parent lineage  $m(k)$ . A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \quad (6)$$

with routing distribution

$$p(m | x) = \text{softmax}(c/\tau)_m, \quad (7)$$

where  $\tau = 1$ .

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, Soft-cascade Transformer uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha p(m(k) | x) e_{m(k),k}(z) + \gamma \log(\epsilon + (1 - \epsilon)p(m(k) | x)), \quad (8)$$

Here  $g(z)$  is a global subtype head,  $e_{m,k}(z)$  is the expert associated with lineage  $m$ ,  $\alpha = 1.0$ ,  $\epsilon = 0.1$ , and  $\gamma$  is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg \max_k s_k(x), \quad \hat{m} = \arg \max_m c_m. \quad (9)$$

#### B.3 CROSS-DISEASE CONTRASTIVE REGULARIZATION

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\text{Proj}(z)}{\|\text{Proj}(z)\|_2} \quad (10)$$

be a normalized projection of the encoder representation. For anchor cell  $i$ , positives are cells with the same subtype but a different disease label. Same-subtype cells from the same disease are excluded by construction. Cells from different subtypes are treated as negatives, with additional emphasis on negatives that share the same coarse lineage. The contrastive loss weight is ramped from 0 to 0.2 after the early warmup period. Because we do not compare this construction with standard supervised contrastive learning across all five folds, we make no claim that the cross-disease positive rule independently improves generalization.

#### B.4 OPTIMIZATION AND MODEL SCOPE

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}} \mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}} \mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t) \mathcal{L}_{\text{contrast}}, \quad (11)$$

where  $\mathcal{L}_{\text{fine}}$  and  $\mathcal{L}_{\text{coarse}}$  are cross-entropy losses with  $\lambda_{\text{fine}} = 1.0$  and  $\lambda_{\text{coarse}} = 0.5$ . The contrastive weight increases to 0.2 according to the schedule in §B.3.

We optimize the cascade using AdamW with learning rate  $10^{-4}$  and weight decay  $10^{-5}$ , gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling because it underperformed natural sampling after matching the number of optimization steps; the single-split development diagnostic is provided in Appendix C.22.

**Backbone-matched Transformer references.** The flat FT-style Transformer reference (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by Soft-cascade Transformer, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same  $M + K$  lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di Montesano et al., 2026), let  $R_{ij} = 1$  when node  $j$  is node  $i$  or its descendant,  $p = \text{softmax}(q)$  over all nodes, and  $s = Rp$ . We minimize  $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$  for observed subtype  $k$  and lineage  $m$ . Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage. Table 3 makes the remaining objective differences explicit. In particular, only the soft-cascade reference uses disease labels to construct cross-disease contrastive pairs; Flat and HCE use no disease labels and no contrastive term. The ladder therefore holds backbone capacity, optimization budget, folds, and checkpoint selection, but it is not an objective-matched component ablation. We do not attribute any score difference to soft routing, hierarchical loss, or contrastive regularization in isolation.

| Component                                         | Flat FT-style Transformer | FT-style Transformer + HCE | Soft-cascade Transformer |
|---------------------------------------------------|---------------------------|----------------------------|--------------------------|
| APT tokenizer and Transformer dimensions          | Same                      | Same                       | Same                     |
| Patient folds, optimizer, budget, checkpoint rule | Same                      | Same                       | Same                     |
| Fine-label supervision                            | Independent CE            | Leaf HCE NLL               | CE                       |
| Coarse-label supervision                          | Independent CE            | Ancestor HCE NLL           | CE                       |
| Cross-disease supervised contrastive term         | No                        | No                         | Yes ( $w \leq 0.2$ )     |
| Disease labels used by cell model                 | No                        | No                         | Contrastive pairing only |
| Global subtype path and lineage experts           | No                        | No                         | Yes                      |
| Soft lineage routing                              | No                        | No                         | Yes                      |

Table 3: Components of the three neural references. “Same” denotes an identical specification, not shared fitted weights. The models match the tokenizer, encoder dimensions, data folds, optimizer, maximum epochs, early stopping rule, and checkpoint criterion, but they do not match the complete training objective. In particular, disease labels enter only the soft-cascade contrastive pairing rule. Consequently, comparisons among these rows evaluate complete recipes and are not component ablations.

Disease recognition is not handled by a dedicated head inside Soft-cascade Transformer. The reference recipe addresses cell-level hierarchy prediction; disease-source prediction is evaluated separately as a within-cohort audit. High disease-source scores do not establish biological attribution or external clinical validity.

**Strictly nested cell-output transfer audit.** For each disease outer fold  $f$ , we train a disease-label-free XGBoost subtype classifier on all 32 outer-development patients and predict the cells of the eight untouched patients in  $f$ . Disease-training features require a second cross-fit: for each of

the other four fixed folds  $g$ , a separate subtype model is trained on the 24 patients outside  $f \cup g$  and predicts fold  $g$ . We average the resulting 27-class probability vectors within patient to obtain soft subtype composition; soft lineage composition sums subtype probabilities under the frozen subtype-to-lineage map. A standardized, class-weighted multinomial LR is fit to the 32 inner-OOF patient vectors, with  $C \in \{0.01, 0.1, 1, 10, 100\}$  selected by patient-level inner CV, and evaluated once on the eight outer-test vectors. Raw proportions are primary; CLR with a prespecified  $10^{-6}$  pseudocount is a sensitivity analysis. All uncertainty and permutation analyses resample or shuffle patients, never cells. To test conditional rather than marginal subtype value, we additionally compare three nested feature sets under the same folds and inner selection: recorded-QC summaries alone, QC plus the five predicted lineage proportions, and QC plus the 27 predicted subtype proportions. The last representation contains lineage implicitly because child probabilities sum to their parents.

The main cell-identity results are reported in Table 1; the following sections supply their implementation and diagnostic details.

## C DATASET AND EVALUATION DETAILS

### C.1 DATASET CONSTRUCTION AND DOCUMENTATION CHECKLIST

The available study workflow establishes the high-level sequence used to construct APT-Bench: clinical peripheral-blood specimens from five cancer groups and healthy controls are converted to single-cell suspensions; cell-profile and viability quality control precedes a paired single-cell transcriptome and 293-aptamer assay; separate transcriptome and aptamer-barcode libraries are linked at the cell level; and transcriptome-derived clusters and annotations provide the cell labels. For a benchmark release, this overview must be accompanied by the following provenance and protocol fields.

**Outstanding disease-confounding validation.** TODO: Obtain a sample-linked acquisition manifest from the data owner: patient and sample IDs, collection and processing dates, APT assay batch, plate/pool, reagent lot, library-preparation batch, sequencing run, site, and available fresh/frozen and handling records. Audit disease-by-batch overlap before choosing a validation design. Where disease groups have adequate cross-batch support, evaluate patient-disjoint, held-out-batch generalization with all preprocessing and selection restricted to training data. If disease and batch are inseparable, obtain crossed-batch samples or an independent APT cohort; statistical adjustment alone cannot identify their separate contributions. The NC matrices must not be treated as external validation if they are the healthy subset of this cohort. These validations are pending, not completed. Expression and QC matrices alone do not establish acquisition provenance; current disease results remain within-cohort associations.

**Transcriptomic preprocessing and cell quality control.** The transcriptomic count matrix was processed with Seurat following its standard workflow of normalization, feature selection, dimensionality reduction, neighborhood-graph construction, and clustering (Hao et al., 2021). DoubletFinder was used to identify and remove predicted doublets (McGinnis et al., 2019). Low-quality cells were additionally filtered using per-cell UMI count, detected-gene count, and mitochondrial-transcript fraction. These steps preceded cluster annotation and construction of the linked APT benchmark cohort. TODO: Report the Seurat and DoubletFinder versions, every parameter and random seed, the exact UMI/gene/mitochondrial thresholds, and cell counts before and after each filter.

**Manual annotation and reference-label reliability.** After Seurat clustering, investigators manually assigned lineage and subtype names by inspecting canonical marker-gene expression reported in prior PBMC and pan-cancer studies (Zhang et al., 2024; Yang et al., 2024). No quantitative annotation confidence, inter-annotator agreement, or external reference-transfer score is available. Accordingly, these labels are operational reference annotations, not error-free biological ground truth, and benchmark scores are conditional on this manual taxonomy. The *Unknown* label was assigned when a cluster did not show a coherent canonical marker pattern for the expected PBMC subtypes; this was a manual reject decision rather than a calibrated confidence threshold. Unknown cells are excluded from the primary 27-subtype benchmark. TODO: Provide the full subtype-by-

marker citation table, the number and expertise of annotators, whether they were blinded to disease, the adjudication procedure, and the explicit rule used to assign Unknown.

**Cross-disease semantics and hierarchy mapping.** The same 27 subtype labels and definitions were applied across all six disease groups; no subtype was renamed or redefined for a particular disease. This gives the labels a consistent operational meaning, but it does not prove that marker expression or biological state is invariant across diseases. Each subtype is assigned to one of five parent lineages by a manually curated lookup. The exported lookup and structural audit are available under `benchmark/release_audit/`: the existing subtype summary has 27 unique subtypes, five parents, and 361,792 cells, and the fold manifest has five disjoint groups of eight patients. These checks establish internal bookkeeping consistency, not independent biological validity. **TODO: Document independent validation of the subtype-to-lineage mapping, including reviewers, disagreements, adjudication, and any marker-based checks.**

## C.2 PAIRED-INPUT INTERNAL RECONSTRUCTION REFERENCE

We use paired RNA as an internal label-reconstruction reference, separate from the primary APT-only model comparison. The current matched-input experiment replaces the earlier data-development HVG representation: 2,000 features are selected by mean-binned log-count dispersion within each inner or outer training subset, using the original RNA count matrix. All 361,792 benchmark cells are aligned by barcode. This controls feature-selection exposure to evaluation patients, but the transcriptomic modality also underlies the manual annotations. Stronger RNA reconstruction therefore does not independently validate those annotations or establish an intrinsic APT information limit.

Within each of the five fixed patient-disjoint folds, we compare five inputs: APT only, paired RNA only, APT+RNA, RNA plus random 293-dimensional noise, and RNA plus APT vectors shuffled within patient. The noise control checks feature dimension, while the within-patient shuffle preserves patient-level APT structure but breaks cell-level RNA-APT pairing. All inputs use the identical deterministic sample of up to 1,000 training cells per development patient and all cells from the eight untouched test patients. RNA features are selected as 2,000 highly variable genes using only the current training fold; all features are standardized using training-fold statistics. We fit patient- and class-weighted linear SGD and one-hidden-layer MLP probes for 40 epochs at both resolutions, using one inner patient-disjoint validation fold to select the regularization strength. Scores use subject-normalized OOF confusion matrices, and 2,000 paired patient-bootstrap replicates quantify modality contrasts. This matched-budget probe uses the main subject-balanced metric, but is not an architecture comparison with the full-training model ladder in Table 1.

| Probe | Input              | Coarse Macro-F1      | Fine Macro-F1        |
|-------|--------------------|----------------------|----------------------|
| SGD   | APT only           | 0.277 [0.254, 0.296] | 0.090 [0.076, 0.098] |
| SGD   | RNA only           | 0.742 [0.731, 0.750] | 0.431 [0.387, 0.453] |
| SGD   | APT + RNA          | 0.751 [0.740, 0.762] | 0.458 [0.413, 0.483] |
| SGD   | RNA + noise        | 0.742 [0.732, 0.751] | 0.409 [0.371, 0.430] |
| SGD   | RNA + shuffled APT | 0.747 [0.732, 0.760] | 0.448 [0.405, 0.471] |
| MLP   | APT only           | 0.310 [0.292, 0.325] | 0.094 [0.081, 0.103] |
| MLP   | RNA only           | 0.871 [0.854, 0.883] | 0.584 [0.529, 0.610] |
| MLP   | APT + RNA          | 0.873 [0.858, 0.884] | 0.606 [0.551, 0.635] |
| MLP   | RNA + noise        | 0.870 [0.855, 0.880] | 0.571 [0.525, 0.593] |
| MLP   | RNA + shuffled APT | 0.876 [0.862, 0.887] | 0.596 [0.544, 0.622] |

Table 4: Paired-input internal reconstruction diagnostic under the five fixed patient-disjoint folds. Each input uses the same deterministic sample of up to 1,000 training cells per development patient, training-fold standardization, all held-out cells, patient- and class-weighted training, and one inner patient-disjoint validation fold for regularization. Brackets are 2,000 paired patient-bootstrap intervals for subject-balanced pooled Macro-F1. RNA features are selected separately within inner and outer training subsets. This prevents feature-selection exposure to their evaluation patients, but the RNA-derived operational annotations are not independently validated by RNA reconstruction.

For the MLP probe, paired RNA improves coarse/fine macro-F1 over APT by +0.562 and +0.490. The concatenated representation improves fine macro-F1 over RNA alone by +0.022 ([0.013, 0.037]), and exceeds RNA plus shuffled APT by +0.010 ([0.002, 0.017]). The linear SGD probe shows the same direction for RNA+APT versus RNA (+0.028; [0.013, 0.041]), but its shuffled-APT contrast is unresolved (+0.010; [-0.005, 0.025]). RNA therefore reconstructs the operational taxonomy far better than APT, while the small combined-input gain should be read as internal complementary association rather than an estimate of annotation accuracy or a clinical claim.

- **TODO: Report recruiting site(s) and dates, prospective versus retrospective collection, exact patient count per group, inclusion/exclusion and diagnostic criteria, and whether each participant contributed exactly one specimen.**
- **TODO: Report available demographics and clinical covariates, including age, sex, disease stage, treatment status, and comorbidities; summarize missingness and state which fields cannot be released.**
- **TODO: Report blood volume, collection tubes, time to processing, cell-isolation and cryopreservation procedures, quality-control thresholds, and sample/cell exclusion counts.**
- **TODO: Report assay/platform and reagent versions, sequencing instruments and depth, 293-probe panel design, and a verified protein-target manifest or explicitly state why only anonymized probe identifiers can be released.**
- **TODO: Report barcode-matching rates, APT normalization and filtering, all remaining software versions and parameters, and whether annotation was performed before investigators examined APT profiles or disease labels.**
- The documented final cell-flow accounting is 374,854 linked post-QC cells – 13,062 Unknown cells = 361,792 benchmark cells. **TODO: Data owners: provide participant and cell counts at all earlier collection/QC exclusions before a complete cohort-flow diagram can be drawn. Supply the approval, consent, access, and licensing evidence listed in the Ethics and Reproducibility Statements.**

### C.3 CHANCE-ADJUSTED RESOLUTION DIAGNOSTIC

Raw coarse and fine Macro-F1 compare five-class and 27-class tasks with different label frequencies and decision boundaries. We therefore treat their difference as descriptive, not evidence of a resolution deficit beyond task cardinality. Using the released 40 patient-specific OOF confusion matrices from the matched paired-input diagnostic, we normalize each matrix to unit mass and average over patients to obtain  $M$ . Let  $p_k = \sum_j M_{kj}$  and  $q_k = \sum_i M_{ik}$ . The independence reference  $M_{ij}^0 = p_i q_j$  retains both true and predicted marginals while breaking their association. We report

$$S_0 = \frac{1}{K} \sum_k \frac{2p_k q_k}{p_k + q_k}, \quad S_{\text{adj}} = \frac{\text{MacroF1}(M) - S_0}{1 - S_0}, \quad (12)$$

with zero contribution when  $p_k + q_k = 0$ . This is F1 of an independence matrix, not the exact expectation of finite-sample permutation F1. Each of 2,000 paired patient-bootstrap draws recomputes  $M$ , both marginals, and the adjustment; the same patient draw is used for both resolutions and all modalities. These intervals condition on the fitted models and do not include retraining variation.

For APT, the MLP raw coarse/fine scores 0.310/0.094 have chance references 0.186/0.029 and adjusted scores 0.152/0.067. The difference shrinks from 0.216 to 0.085 (95% interval [0.068, 0.104]). The corresponding SGD adjusted gap is 0.054 ([0.034, 0.078]). Adjusted RNA and APT+RNA differences remain larger for both probes, around 0.25–0.27. The adjustment accounts for marginal chance agreement, but does not control label boundaries, class count, annotation reliability, or within-lineage separability; none of these residual differences establishes a cardinality-independent biological granularity effect.

True-lineage-conditioned subtype decoding is not evaluated here. It requires OOF subtype scores restricted to the children of the true lineage (or separately trained conditional classifiers); hard predictions and confusion matrices cannot recover that decision. In particular, filtering to predictions that already match the true lineage would select easier cases rather than provide this control. Thus the present diagnostic cannot attribute fine errors uniquely to lineage identification versus within-lineage discrimination.

| Probe | Input     | Resolution | Raw F1 | Chance ref. | Adjusted F1 [95% interval] |
|-------|-----------|------------|--------|-------------|----------------------------|
| SGD   | APT       | coarse     | 0.277  | 0.185       | 0.113 [0.093, 0.134]       |
| SGD   | APT       | fine       | 0.090  | 0.033       | 0.059 [0.046, 0.067]       |
| SGD   | RNA       | coarse     | 0.742  | 0.197       | 0.678 [0.666, 0.689]       |
| SGD   | RNA       | fine       | 0.431  | 0.035       | 0.410 [0.367, 0.433]       |
| SGD   | APT + RNA | coarse     | 0.751  | 0.198       | 0.690 [0.676, 0.702]       |
| SGD   | APT + RNA | fine       | 0.458  | 0.035       | 0.438 [0.393, 0.463]       |
| MLP   | APT       | coarse     | 0.310  | 0.186       | 0.152 [0.134, 0.171]       |
| MLP   | APT       | fine       | 0.094  | 0.029       | 0.067 [0.055, 0.076]       |
| MLP   | RNA       | coarse     | 0.871  | 0.200       | 0.839 [0.819, 0.854]       |
| MLP   | RNA       | fine       | 0.584  | 0.037       | 0.568 [0.512, 0.596]       |
| MLP   | APT + RNA | coarse     | 0.873  | 0.200       | 0.842 [0.824, 0.855]       |
| MLP   | APT + RNA | fine       | 0.606  | 0.037       | 0.591 [0.535, 0.621]       |

Table 5: Marginal-chance adjustment for the paired-input reconstructability diagnostic. Scores use subject-balanced pooled confusion matrices from the same 40 OOF patients. Each of 2,000 paired patient-bootstrap draws recomputes the observed score and null marginals. This is an association-breaking reference, not a task-cardinality, annotation-quality, or within-lineage difficulty control.

#### C.4 SEQUENTIAL CONTROLS: SYNTHETIC MECHANISM CHECK

The real-data class-matched sampler fixes class quotas and spreads cells across donors within each class. To distinguish these operations, we implement four conditions with common random cell priorities and nested development subject sets: (A) uniform pooled-cell sampling; (B) restriction to the eight-subject reference class set, reserving one cell per class before uniform filling; (C) exact reference class quotas with uniform within-class sampling; and (D) the same quotas with capacity-constrained donor round robin in randomized donor order. At fixed total, exact class proportions and exact counts are equivalent constraints, not separate ablations. This A condition is not the legacy equal-cell-per-subject sampler. Sequential contrasts are order-dependent descriptions, not causal mediation estimates.

We evaluate this implementation on synthetic data, not a semi-synthetic clinical cohort. Each of 20 seeds generates six class centers  $\mu_k \sim \mathcal{N}(0, s^2 I_{12})$  and 48 subjects with 400 cells each. Subject label probabilities are  $\text{softmax}(\eta_p)$ , where  $\eta_p \sim \mathcal{N}(0, m^2 I_6)$ , and features are  $x_{pi} = \mu_{y_{pi}} + b_p + \epsilon_{pi}$ , with  $b_p \sim \mathcal{N}(0, a^2 I_{12})$  and  $\epsilon_{pi} \sim \mathcal{N}(0, I_{12})$  independently. We vary  $a, m \in \{0, 1.5\}$  and  $s \in \{0.5, 1.5\}$ . The first 32 subjects form the development pool; the remaining 16 are untouched test subjects. An unweighted LR probe is refit with training-only standardization for each condition and  $P \in \{8, 32\}$  at total 1,600 training cells. All conditions reuse the same realization and test subjects within a seed.

With exchangeable subjects ( $a = m = 0$ ), mean effects are near zero. With label mixture variation alone and  $s = 0.5$ , the unmatched mean effect is +0.0245, compared with -0.0011 after quota matching. Conversely, with subject offsets alone and  $s = 0.5$ , quota matching retains a mean effect of +0.0280, compared with unmatched +0.0281. Thus matching does not invariably eliminate a breadth benefit in this generator. Seed variability is substantial; these results do not establish interval coverage, a universal correction, or the mechanism in APT, COMBAT, or OneK1K. Support is complete in these simulations, so they do not test missing-class recovery. Separate real-cohort A–D results are not included in this version; the existing class-matched tables must not be interpreted as that decomposition.

#### C.5 KNOWN-NUL AND INDEPENDENT-TARGET VALIDATION

Known generator parameters are not themselves a known learning effect. We therefore define the target explicitly as

$$\theta_C = \mathbb{E}[S_C(32, Q_8) - S_C(8, Q_8)], \quad (13)$$

where  $S$  is subject-balanced pooled six-subtype Macro-F1 on untouched test subjects, and  $Q_8$  is the development-only eight-subject class-quota reference. The expectation integrates over generated cohorts and sampling. This overlap-quota policy target excludes gains from acquiring labels absent at  $P = 8$ ; it is not the total practical value of recruiting more subjects.

| Offset | Mixture | Separation | A: Unmatched    | C: Quota        | D: Quota + donor |
|--------|---------|------------|-----------------|-----------------|------------------|
|        |         |            | +0.001          | +0.001          | +0.001           |
| 0.0    | 0.0     | 0.5        | [-0.007,+0.007] | [-0.007,+0.008] | [-0.006,+0.007]  |
|        |         |            | -0.000          | -0.000          | -0.000           |
| 0.0    | 0.0     | 1.5        | [-0.003,+0.001] | [-0.003,+0.002] | [-0.003,+0.001]  |
|        |         |            | +0.024          | -0.001          | -0.002           |
| 0.0    | 1.5     | 0.5        | [-0.005,+0.075] | [-0.012,+0.010] | [-0.012,+0.007]  |
|        |         |            | +0.002          | +0.000          | -0.000           |
| 0.0    | 1.5     | 1.5        | [-0.005,+0.009] | [-0.005,+0.003] | [-0.003,+0.005]  |
|        |         |            | +0.028          | +0.028          | +0.031           |
| 1.5    | 0.0     | 0.5        | [-0.028,+0.103] | [-0.023,+0.103] | [-0.015,+0.104]  |
|        |         |            | +0.043          | +0.042          | +0.042           |
| 1.5    | 0.0     | 1.5        | [-0.005,+0.114] | [-0.006,+0.113] | [-0.009,+0.115]  |
|        |         |            | +0.027          | +0.016          | +0.025           |
| 1.5    | 1.5     | 0.5        | [-0.051,+0.133] | [-0.059,+0.115] | [-0.039,+0.103]  |
|        |         |            | +0.040          | +0.029          | +0.048           |
| 1.5    | 1.5     | 1.5        | [-0.029,+0.130] | [-0.054,+0.114] | [-0.008,+0.136]  |

Table 6: Synthetic mechanism check: subject-balanced Macro-F1 difference between 32 and 8 training subjects at fixed total 1,600 cells. Entries are means across 20 independent seeds with empirical 2.5th–97.5th seed percentiles, not confidence intervals of the mean. Condition B equals A in every run because support is already complete and the support floor does not alter selection. All eight parameter settings are reported.

We generate a hierarchy of three lineages with two sibling subtypes each in eight dimensions. Fixed class centers are  $\mu_k = 2e_{\lfloor k/2 \rfloor} + (-1)^{k+1} se_{3+\lfloor k/2 \rfloor}$  for  $k = 0, \dots, 5$  (zero-indexed axes). Features follow  $x_{pi} = \mu_{y_{pi}} + b_p + u_{p,g(i)} + v_{h(p)} + \epsilon_{pi}$ , with independent isotropic Gaussian effects: donor standard deviation  $a$ , ten-cell block standard deviation  $c$ , four-donor batch standard deviation  $b$ , and residual standard deviation 1. Within-subject correlation can thus arise at donor or cell-block level; these covariance components should not be interpreted as independent empirical estimates of biological heterogeneity. Subject label probabilities are softmax-normal with logit standard deviation 1.5, and the sixth subtype is available in only 10% of subjects. Each cohort contains 32 development and 40 test subjects, with 120 cells per subject. The training total is 480 at both  $P = 8$  and  $P = 32$ ; preprocessing and LR fitting follow the preceding synthetic probe. Train and test subjects and batches are disjoint. All tasks retain the fixed six-class scoring ontology.

The base setting is  $(a, c, b, s) = (0, 0, 0, 0.8)$ . Conditional on the label-only quota reference, training features in each class are iid with the same distribution for both subject budgets. Consequently, the C fitted-model distribution, and hence expected test score, is identical:  $\theta_C = 0$  exactly. This argument does not require balanced natural class proportions or complete support. Additional settings change only  $a$  to 0.5 or 1.5; stress settings start from  $a = 0.5$  and set  $c = 0.7$ ,  $s = 0.3$ , or  $b = 1$ , respectively. In the batch setting, increased subject breadth also increases batch coverage; the target remains a policy contrast rather than a separate donor effect.

For every non-null setting we estimate  $\theta_C$  using 512 independent reference cohorts, with a direct uniform-without-replacement index sampler separate from the evaluated priority-sampling implementation. Another 100 independent cohorts evaluate A/B/C using paired subject pools, cell priorities, and test sets. Reference and evaluation random-seed ranges are disjoint. We report reference Monte Carlo standard errors (MCSE), estimator bias, and RMSE. Normal Monte Carlo intervals for bias combine evaluation-mean variance and independent reference-mean variance; these are not patient-bootstrap coverage measurements or multiplicity-adjusted tests.

Under the exact null, A and B yield mean effects +0.0202 and +0.0119, whereas C yields +0.0004 (bias interval  $[-0.0017, 0.0026]$ ). The independent reference check is +0.00043 with MCSE 0.00053, consistent with the analytic null. Mild and strong donor shifts have reference effects +0.0110 and +0.0135; C estimates +0.0102 and +0.0168, respectively, rather than mechanically forcing both to zero. Higher shift variance does not guarantee a proportionally larger learning benefit. Across all six settings, C has lower RMSE against this target than A or B. Nevertheless, A/B differences can be legitimate benefits of their own allocation policies: their error relative to  $\theta_C$  demonstrates estimand mismatch, not that all observed breadth gains are spurious. Non-null targets

| Setting            | Target (MCSE)   | A: mean / RMSE   | B: mean / RMSE   | C: mean / RMSE   |
|--------------------|-----------------|------------------|------------------|------------------|
| Known null         | 0 (exact)       | +0.0202 / 0.0354 | +0.0119 / 0.0249 | +0.0004 / 0.0109 |
| Mild donor shift   | 0.0110 (0.0009) | +0.0278 / 0.0341 | +0.0221 / 0.0264 | +0.0102 / 0.0191 |
| Strong donor shift | 0.0135 (0.0015) | +0.0332 / 0.0412 | +0.0316 / 0.0392 | +0.0168 / 0.0358 |
| Correlated cells   | 0.0108 (0.0008) | +0.0301 / 0.0334 | +0.0264 / 0.0290 | +0.0102 / 0.0199 |
| Weak subtypes      | 0.0049 (0.0010) | +0.0247 / 0.0361 | +0.0235 / 0.0346 | +0.0016 / 0.0222 |
| Batch shift        | 0.0181 (0.0017) | +0.0318 / 0.0463 | +0.0299 / 0.0438 | +0.0192 / 0.0429 |

Table 7: Known-null and independent Monte Carlo validation of the overlap-quota target. A, B, and C denote unmatched, support-matched, and exact-quota sampling. Non-null targets use 512 independent reference cohorts; estimator means and RMSE use 100 separate evaluation cohorts. MCSE denotes reference Monte Carlo standard error. The target is the expected effect of the C allocation policy, not unrestricted subject recruitment. All six settings are reported; RMSE is measured relative to the exact null or estimated reference target.

remain Monte Carlo approximations. This validation neither identifies real-cohort causal mediation nor establishes coverage of the joint uncertainty procedure used on real data.

## C.6 BENCHMARK INTERFACE AND RELEASE CONTRACT

The primary hierarchy submission interface contains pseudonymous cell and patient IDs, true and predicted lineage, and true and predicted subtype. The disease interface contains one row per patient with the true disease and six class scores. The secondary budget diagnostic additionally records the aptamer measurement budget, inference-time cell budget, selection seed, and the development-only feature ranking. These common interfaces permit new models to be scored without reproducing their internal training pipelines.

The evaluation layer is portable to other clinical single-cell datasets. It implements group-disjoint out-of-fold prediction; patient-clustered bootstrap uncertainty; subject-balanced and cell-weighted pooled coarse and fine Macro-F1, exact-path accuracy, root-excluded hierarchical F1, subtype-tree distance, the descriptive coarse-minus-fine difference, and sibling versus cross-lineage error; patient-level aggregation versus cell-level pseudoreplication; patient-centering and identity diagnostics; and nested feature- and cell-budget curves. A future dataset can replace the assay features, hierarchy, or clinical endpoint while retaining these interfaces. This modular task–dataset–metric separation follows recent living single-cell benchmarks (Luecken et al., 2025); standardized metadata and programmatic access similarly enable cross-study reuse in CZ CELLxGENE (CZI Cell Science Program et al., 2025).

A complete release must contain the processed APT matrix; pseudonymous cell and patient identifiers; disease and cell labels; hierarchy and feature metadata; the immutable fold manifest; metric implementations; baseline configurations; and a single command that regenerates the headline tables from model predictions. The fold manifest, exported hierarchy, selected check-sums, and artifact-rebuild instructions are now recorded in `benchmark/release_audit/` and `REPRODUCIBILITY.md`; these cover selected results, not the complete dataset or training pipeline. **TODO: Before submission, finalize anonymous data/code access, explicit licenses, original training environment locks, an archival DOI, and the end-to-end command. If participant-level data require controlled access, release the complete schema, all labels permitted by consent, the split manifest, evaluator, and access procedure.**

## C.7 CONFOUNDING AUDIT OF WITHIN-COHORT DISEASE ASSOCIATION

**Exploratory scope and independent sample size.** All disease analyses reuse one cohort of 40 independent patients across six groups (6–8 patients per group); the 361,792 cells are repeated observations, not additional independent disease outcomes. Outer-fold disease models are developed on only 32 patients. Nested cross-fitting separates training from evaluation but does not increase the number of independent subjects, remove unmeasured confounding, or turn repeated analyses into independent replications. Regularization and inner validation mitigate model-selection risk without eliminating the uncertainty associated with this small development sample. The representation com-

| (a) Task and evaluation scope |               |                             |                |               |
|-------------------------------|---------------|-----------------------------|----------------|---------------|
| Resource                      | Primary input | Reported scope              | Main unit      | Hierarchy     |
| Open Problems                 | Multiple      | 12 living tasks             | Task-specific  | Task-specific |
| scMultiBench                  | RNA/ADT/ATAC  | 64 real + 22 simulated      | Cell/dataset   | No            |
| SPARE                         | scRNA-seq     | 5 datasets, 8 methods       | Sample         | No            |
| PaSCient                      | scRNA-seq     | 12.5M cells, 2,700 patients | Patient        | No            |
| APT-Bench                     | 293 aptamers  | 1 cohort, 40 patients       | Cell + patient | Yes           |

| (b) Protocol components and availability |               |                |                   |                 |
|------------------------------------------|---------------|----------------|-------------------|-----------------|
| Resource                                 | Patient task  | Two-axis scale | Assay budget      | Public pipeline |
| Open Problems                            | Task-specific | No             | No                | Yes             |
| scMultiBench                             | No            | No             | Feature selection | Yes             |
| SPARE                                    | Yes           | No             | No                | Yes             |
| PaSCient                                 | Yes           | No             | No                | Not audited     |
| APT-Bench                                | Audit         | Yes            | Yes               | Pending release |

Table 8: Scope comparison with recent single-cell evaluation resources (Luecken et al., 2025; Liu et al., 2025; Shitov et al., 2025; Liu et al., 2026). “Task-specific” means that the living platform defines multiple contracts rather than one shared clinical protocol. APT-Bench is narrower in cohorts but combines a fixed aptamer modality, patient-disjoint coarse-to-fine cell typing, within-cohort patient-level audit, and controlled patient/cell and assay budgets. Its public release remains a submission requirement, not a completed claim.

| Aggregation / Representation                      | Patient Acc. | Macro-F1 | Macro-AUROC | Perm. $p$ |
|---------------------------------------------------|--------------|----------|-------------|-----------|
| Log cell count + LR <sup>‡</sup>                  | 0.575        | 0.531    | 0.800       | 0.00010   |
| All recorded QC proxies + LR <sup>¶</sup>         | 0.550        | 0.566    | 0.859       | –         |
| Nested soft subtype composition + LR <sup>§</sup> | 0.600        | 0.593    | 0.854       | 0.00010   |
| Nested soft lineage composition + LR <sup>§</sup> | 0.475        | 0.477    | 0.766       | 0.00020   |
| Reference subtype composition + LR <sup>‡</sup>   | 0.500        | 0.516    | 0.776       | 0.00010   |
| Reference lineage composition + LR <sup>‡</sup>   | 0.325        | 0.301    | 0.743       | 0.0184    |

<sup>‡</sup> Patient-level technical or paired-transcriptomic oracle features. <sup>¶</sup> Recovered cell count plus five patient summaries of recorded RNA UMI and detected-gene QC fields; see Appendix C.7. <sup>§</sup> Soft probabilities from disease-label-free XGBoost under strict nested cross-fitting. Composition rows use raw proportions; LR regularization is tuned only within each outer-development set. Composition permutation tests use 10,000 patient-label shuffles.

Table 9: **Associational patient-characterization audit** ( $n = 40$ ). Predicted composition retains disease-associated variation, but its gains over lineage composition, recovered cell count, and recorded RNA-QC proxies have patient-bootstrap intervals spanning zero. These representations may contain biological, patient, or technical structure. Full direct-APT settings and confounding controls appear in Appendix C.7; no row establishes biological mediation or clinical generalization.

parisons and post-hoc sensitivities below are exploratory, not confirmatory evidence of incremental subtype value or clinical utility. Reported patient-bootstrap intervals do not account for the full adaptive history of analysis choices. Clinical validation would require an independent APT cohort with recorded acquisition covariates and a frozen preprocessing, feature-selection, and prediction pipeline; the external scaling experiments do not supply that disease-validation cohort.

Table 9 in the main text summarizes the patient-representation comparison; this section supplies its controls and interpretation limits.

**Cell-level prediction of the disease source of a held-out patient.** Table 9 reports the exploratory patient-characterization audit. For completeness, Table 11 reports the pooled *cell-level* source-prediction metric, in which every held-out cell is treated as a target and pooled across all 361,792 cells. This is an easier and statistically different task than patient-level characterization, since cells

from the same patient are correlated and patients with more cells contribute more weight. We report it only as a confounding diagnostic.

**Non-identifiability of disease and acquisition effects.** Every participant contributes one acquisition, so patient identity is nested within both disease group and any unrecorded processing batch. The released metadata contain disease and patient identifiers plus transcriptomic UMI and detected-gene counts, but no plate, run, date, library, operator, or site field. Consequently, neither patient-disjoint prediction nor label permutation can identify whether a patient-associated APT offset is biological, technical, or mixed. No analysis below is interpreted as disease biology, biomarker validation, or clinical performance.

**Recorded nuisance-only baselines.** We aggregate the only available per-cell QC fields to each patient: recovered cell count; medians of  $\log(1 + \text{UMI})$  and  $\log(1 + \text{detected genes})$ ; median detected-gene/UMI ratio; and the interquartile ranges of the two log-count variables. Each feature set uses the same outer folds as the composition bridge, four-fold inner patient CV on the 32 outer-development patients to select LR regularization, refitting on all 32, and one evaluation on the sealed eight patients. Table 10 shows that these recorded QC proxies alone approach the predicted-composition result. They remain proxies rather than measured batch labels, so this analysis detects known nuisance structure but cannot rule out unrecorded acquisition effects.

We also test conditional subtype value rather than comparing each representation in isolation. Under the same nested folds and inner regularization selection, recorded QC plus predicted lineage reaches macro-F1 0.583 and AUROC 0.875; recorded QC plus the full predicted subtype vector reaches 0.599 and 0.825. The subtype-minus-lineage macro-F1 increment is +0.016 with 95% paired patient-bootstrap CI  $[-0.120, 0.169]$  (10,000 resamples). Thus subtype detail does not provide a resolved conditional increment in this cohort.

| Recorded feature set               | $d$ | Acc.  | Macro-F1 | Macro-AUROC |
|------------------------------------|-----|-------|----------|-------------|
| Recovered cell count               | 1   | 0.575 | 0.531    | 0.800       |
| RNA QC summaries                   | 3   | 0.475 | 0.478    | 0.787       |
| All recorded QC proxies            | 6   | 0.550 | 0.566    | 0.859       |
| Predicted soft subtype composition | 27  | 0.600 | 0.593    | 0.854       |
| Recorded QC + predicted lineage    | 11  | 0.575 | 0.583    | 0.875       |
| Recorded QC + predicted subtype    | 33  | 0.600 | 0.599    | 0.825       |

Table 10: Matched patient-level nuisance audit. All rows use the same outer folds and inner patient-level regularization selection. The predicted-composition minus all-recorded-QC Macro-F1 difference is +0.027 with a 95% paired patient-bootstrap interval of  $[-0.157, 0.215]$  (2,000 resamples). Recorded QC fields are incomplete proxies, not batch labels. In the conditional audit, the subtype-minus-lineage Macro-F1 increment is +0.016 with 95% paired patient-bootstrap interval  $[-0.120, 0.169]$  (10,000 resamples).

**Patient-wise centering sensitivity.** To test whether disease prediction relies on each patient’s absolute expression level rather than only on within-patient relative structure, we repeated Logistic Regression training and evaluation after subtracting each patient’s own per-aptamer median from every cell in both training and test data. This transformation preserves within-patient cell-to-cell variation while removing a patient-specific location shift. Cell-level accuracy drops from 0.991 to 0.401 (macro-F1 0.989  $\rightarrow$  0.348), and patient-level majority-vote accuracy drops from 40/40 to 26/40. This collapse shows that disease recognition depends substantially on patient-level absolute-expression structure, but it does not by itself determine whether that structure is biological, technical, or a mixture of both.

**Patient-fingerprint diagnostic.** We additionally trained a 40-way classifier to predict patient identity from individual cells, using a within-patient 50/50 cell split for this diagnostic only. The classifier identifies the correct patient from a single held-out cell with 88.4% accuracy (macro-F1 0.862), against a 2.5% chance level. Individual cells therefore carry a strong and easily recoverable patient-specific signature, consistent with the centering sensitivity above.

| Model                              | Accuracy     | Macro-F1     | Weighted-F1  | Balanced Acc. |
|------------------------------------|--------------|--------------|--------------|---------------|
| Logistic Regression                | <b>0.991</b> | <b>0.989</b> | <b>0.991</b> | <b>0.989</b>  |
| XGBoost                            | 0.990        | 0.988        | 0.990        | 0.987         |
| Linear SVM                         | 0.988        | 0.986        | 0.988        | 0.985         |
| Random Forest                      | 0.963        | 0.955        | 0.963        | 0.953         |
| Reference Correlation <sup>†</sup> | 0.921        | 0.909        | 0.921        | 0.915         |

<sup>†</sup> SingleR-style reference-correlation classifier (Spearman nearest-centroid mapping), following the template-matching principle used by reference-mapping methods such as SingleR (Aran et al., 2019).

Table 11: **Cell-level** prediction of the disease source of a held-out patient’s cells (6-class), pooled over all 361,792 out-of-fold cells from the patient-level 5-fold protocol. This is a different, easier statistical task than the exploratory patient-level characterization reported in Table 9: cells from the same patient are correlated observations rather than independent draws, and this metric implicitly weights patients with more cells more heavily. We report it here only as a cell-level reference point, not as a patient-level diagnostic result.

**Patient-level summary-feature control.** Finally, to confirm that the disease signal does not rely only on cell-level pseudo-replication, we trained a genuinely patient-level classifier using each patient’s median 293-dimensional APT profile. This summary-feature model reaches patient accuracy 0.950, patient macro-F1 0.948, and macro-AUROC 1.000. Together with the sensitivity analyses above, this indicates that disease information is present at the patient level but strongly entangled with absolute-expression offsets that cannot be cleanly separated without acquisition-batch meta-data.

**Strict nested cell-output transfer audit.** Ordinary pooled OOF predictions are insufficient for stacking because a base model used to create a disease-training feature may have seen the eventual disease outer-test patients. We therefore use the nested protocol defined in Methods: within each outer fold, four XGBoost models trained on 24 patients generate inner-OOF features for the 32 disease-training patients, and a fifth model trained on all 32 predicts the eight outer-test patients. XGBoost receives subtype labels but no disease labels. Soft subtype composition reaches accuracy/macro-F1/AUROC 0.600/0.593/0.854 and soft lineage composition reaches 0.475/0.477/0.766. The paired macro-F1 difference is +0.116 (95% patient-bootstrap CI [−0.015, 0.262]). Gold subtype and lineage compositions reach macro-F1 0.516 and 0.301, respectively. The apparently stronger predicted than gold point estimate is not statistically resolved and warns that predicted probabilities may encode model-confidence or technical structure beyond true cell abundance.

This experiment is therefore a representation-transfer audit, not an error- propagation analysis. It does not establish that the transferred signal originates from subtype abundance, that fine subtype is more useful than coarse lineage, that better typing improves patient characterization, or that the signal is independent of confidence and technical structure. Consistent with this distinction, the soft-minus-hard subtype macro-F1 difference is only +0.069 (95% patient-bootstrap CI [−0.077, 0.221]), and the predicted-minus-gold subtype difference is +0.077 (95% CI [−0.100, 0.256]).

Across 10,000 complete patient-label permutations, predicted subtype and lineage composition have plus-one  $p = 0.00010$  and  $p = 0.00020$ . However, log cell count alone reaches macro-F1 0.531 ( $p = 0.00010$ ), and the predicted-subtype minus cell-count difference is +0.062 (95% CI [−0.087, 0.218]). Equal-cell subsampling at 100, 500, and 2,000 cells per patient preserves the qualitative result over 50 seeds. A 50-repeat global-multinomial composition control averages macro-F1 0.170, and fold-ID prediction from global OOF composition is below chance. These controls support a predictive and associational bridge, not causal mediation or independent biological validation.

## C.8 PATIENT COMPOSITION RECOVERY

For each of the 40 outer-test patients, we compare the mean soft cell-type probabilities with reference cell fractions. The constant baseline is the equal-patient mean reference composition of the 32 outer-development patients in that fold; no test-patient annotations enter its construction. For  $K$  classes

and  $N$  test patients, the recovery error is

$$E = \frac{1}{N} \sum_{p=1}^N \frac{1}{K} \sum_{k=1}^K |\hat{\pi}_{pk} - \pi_{pk}|.$$

We report  $E_{\text{baseline}} - E_{\text{model}}$ , so positive values favor predicted compositions. All five lineages or all 27 subtypes are included, without selecting well-recovered classes. Absolute errors across these two resolutions are not directly comparable because their denominators differ. Using 2,000 disease-stratified paired patient-bootstrap replicates, subtype error decreases from 0.02314 to 0.01478, a reduction of 0.00836 ([0.00648, 0.01010]). Lineage error changes from 0.05495 to 0.05827, a reduction of  $-0.00332$  ( $[-0.01389, 0.00670]$ ). Intervals are pointwise and conditional on the completed cross-fitted cell models; they do not include training-subset or model-fitting uncertainty. The accompanying per-class audit reports error, signed bias, and rank correlation for every class. Disease-adjusted rank correlations residualize both ranked fractions on disease-group indicators and are descriptive, not evidence of biological specificity or causal independence from disease. Reference recovery does not validate the annotation itself or imply improved disease characterization.

| Representation                | Legacy pooled OOF | Strict nested |
|-------------------------------|-------------------|---------------|
| Gold subtype composition      | 0.572             | 0.516         |
| Predicted subtype composition | 0.400             | 0.593         |
| Gold lineage composition      | 0.377             | 0.301         |
| Predicted lineage composition | 0.282             | 0.477         |

Table 12: Protocol audit of patient macro-F1. The legacy predicted rows use hard labels from one global pooled-OOF prediction set, CLR features, and fixed LR regularization; the strict rows use nested soft probabilities, raw proportions, and development-only LR selection. Because both feature construction and downstream fitting changed, the numerical differences are not estimates of a leakage effect. Only strict nested results support current claims.

**Patient-label permutation control.** We evaluated the same patient-level median-profile classifier under 1,000 label permutations. Disease labels were permuted across the 40 patients as indivisible units, preserving the disease-group counts (6, 8, 7, 7, 6, 6); labels were never shuffled across cells. We retained the original five patient folds rather than restratifying after permutation. No permuted dataset reached the observed accuracy of 0.950 or macro-F1 of 0.948, giving plus-one permutation  $p = (0 + 1)/(1000 + 1) = 0.000999$  for both statistics. The null accuracy mean was 0.137, with empirical 95% interval [0.025, 0.275]. This rejects a random patient-label explanation under the fixed protocol, but does not distinguish biological signal from patient-associated technical structure.

## C.9 PER-SUBTYPE SUPPORT AND PERFORMANCE AUDIT

Figure 2 visualizes the two marginal support-F1 associations for Soft-cascade Transformer and XGBoost. The vertical stack at 40 patients makes the coverage ceiling explicit. These plots and the partial correlations are exploratory: they neither establish causality nor test whether two dependent correlations differ significantly.

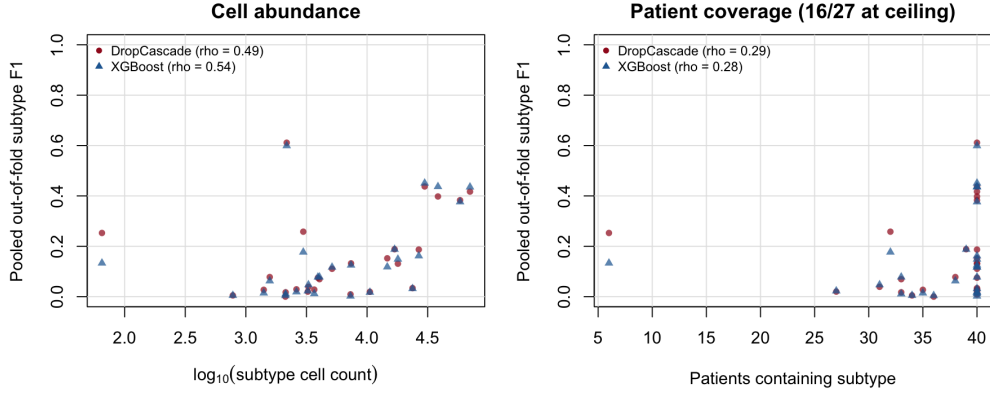


Figure 2: Per-subtype pooled out-of-fold F1 versus cell abundance (left) and patient coverage (right). Each marker is one of 27 subtypes; both models use the same ground-truth support values. Spearman correlations are marginal, and 16 of 27 subtypes occur in all 40 patients.

(a) Support–performance associations

| Model                      | $\rho_c$ | $\rho_p$ | $\rho_{c p}$ | $\rho_{p c}$ |
|----------------------------|----------|----------|--------------|--------------|
| Soft-cascade Transformer   | 0.486    | 0.288    | 0.411        | -0.051       |
| Flat FT-style Transformer  | 0.505    | 0.255    | 0.464        | -0.123       |
| FT-style Transformer + HCE | 0.513    | 0.318    | 0.427        | -0.034       |
| XGBoost                    | 0.540    | 0.275    | 0.496        | -0.130       |
| Logistic Regression        | 0.435    | 0.107    | 0.488        | -0.267       |
| Linear SVM                 | 0.339    | -0.012   | 0.463        | -0.335       |
| Random Forest              | 0.661    | 0.418    | 0.564        | -0.035       |
| Reference Correlation      | 0.376    | 0.003    | 0.499        | -0.354       |

(b) Granularity and error structure

| Model                      | $G$   | $R_{\text{sibling}}$ | Cross-lineage |
|----------------------------|-------|----------------------|---------------|
| Soft-cascade Transformer   | 0.174 | 0.250                | 0.750         |
| Flat FT-style Transformer  | 0.176 | 0.243                | 0.757         |
| FT-style Transformer + HCE | 0.183 | 0.257                | 0.743         |
| XGBoost                    | 0.179 | 0.246                | 0.754         |
| Logistic Regression        | 0.167 | 0.295                | 0.705         |
| Linear SVM                 | 0.200 | 0.265                | 0.735         |
| Random Forest              | 0.158 | 0.191                | 0.809         |
| Reference Correlation      | 0.099 | 0.225                | 0.775         |

Table 13: Exploratory subtype diagnostics on cell-weighted pooled patient-disjoint predictions.  $\rho_c$  and  $\rho_p$  are marginal Spearman associations of subtype F1 with log cell count and patient coverage; partial correlations residualize ranked variables.  $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$ . Among errors,  $R_{\text{sibling}}$  retains the true parent and Cross-lineage is its complement. These cohort-level summaries are descriptive, not causal.

Table 14 gives all subtype-level quantities and 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Counts and coverage use pooled ground truth, while F1 uses the same patient-disjoint OOF predictions as Table 1. Mean cells per patient averages only over patients in which the subtype is observed.

Table 14: Per-subtype support and pooled out-of-fold F1 with 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Cells/patient is the mean among patients in which that subtype is observed. Head and Tail denote the seven most and seven least abundant subtypes; all others are Mid.

| Subtype           | Lineage | Cells  | Patients | Cells/patient | Regime | Soft-cascade F1 [95% CI] | XGB F1 [95% CI]      |
|-------------------|---------|--------|----------|---------------|--------|--------------------------|----------------------|
| Erythrocyte       | Other   | 65     | 6        | 10.8          | Tail   | 0.253 [0.000, 0.395]     | 0.133 [0.000, 0.545] |
| Atypical Memory B | B       | 781    | 34       | 23.0          | Tail   | 0.005 [0.000, 0.014]     | 0.005 [0.000, 0.014] |
| CD4 Monocyte-4    | Myeloid | 1,405  | 35       | 40.1          | Tail   | 0.027 [0.006, 0.053]     | 0.014 [0.000, 0.033] |
| platelet          | Other   | 1,574  | 38       | 41.4          | Tail   | 0.078 [0.016, 0.125]     | 0.063 [0.015, 0.106] |
| Tcm               | T       | 2,121  | 36       | 58.9          | Tail   | 0.000 [0.000, 0.000]     | 0.004 [0.000, 0.016] |
| Memory B-2        | B       | 2,130  | 33       | 64.5          | Tail   | 0.017 [0.004, 0.033]     | 0.011 [0.000, 0.024] |
| Plasma            | B       | 2,173  | 40       | 54.3          | Tail   | 0.611 [0.522, 0.684]     | 0.599 [0.496, 0.685] |
| pDC               | Myeloid | 2,615  | 40       | 65.4          | Mid    | 0.029 [0.010, 0.061]     | 0.019 [0.005, 0.040] |
| CD14 Monocyte-3   | Myeloid | 2,977  | 32       | 93.0          | Mid    | 0.258 [0.170, 0.306]     | 0.177 [0.103, 0.211] |
| CD8 + CTL-3       | T       | 3,244  | 27       | 120.1         | Mid    | 0.020 [0.004, 0.035]     | 0.023 [0.008, 0.043] |
| GCB               | B       | 3,283  | 31       | 105.9         | Mid    | 0.039 [0.007, 0.071]     | 0.047 [0.001, 0.073] |
| Cycling T         | T       | 3,667  | 40       | 91.7          | Mid    | 0.028 [0.015, 0.043]     | 0.012 [0.003, 0.022] |
| CD14 Monocyte-2   | Myeloid | 3,931  | 40       | 98.3          | Mid    | 0.075 [0.046, 0.102]     | 0.078 [0.056, 0.097] |
| CD8 + CTL-2       | T       | 4,049  | 33       | 122.7         | Mid    | 0.069 [0.017, 0.116]     | 0.078 [0.015, 0.113] |
| CD16 Monocyte-2   | Myeloid | 5,142  | 40       | 128.6         | Mid    | 0.111 [0.064, 0.161]     | 0.117 [0.063, 0.164] |
| CD8 + CTL-1       | T       | 7,303  | 40       | 182.6         | Mid    | 0.009 [0.000, 0.024]     | 0.002 [0.001, 0.004] |
| cDC2              | Myeloid | 7,376  | 40       | 184.4         | Mid    | 0.132 [0.099, 0.181]     | 0.125 [0.079, 0.167] |
| CD8 Tem-2         | T       | 10,548 | 40       | 263.7         | Mid    | 0.019 [0.005, 0.038]     | 0.017 [0.008, 0.030] |
| CD16 Monocyte-1   | Myeloid | 14,678 | 40       | 366.9         | Mid    | 0.152 [0.111, 0.198]     | 0.118 [0.090, 0.150] |
| CD4 + Tcm-2       | T       | 16,861 | 39       | 432.3         | Mid    | 0.188 [0.111, 0.236]     | 0.188 [0.136, 0.220] |
| Naive T           | T       | 17,960 | 40       | 449.0         | Head   | 0.131 [0.067, 0.183]     | 0.149 [0.073, 0.199] |
| Memory B-1        | B       | 23,722 | 40       | 593.0         | Head   | 0.034 [0.020, 0.048]     | 0.032 [0.017, 0.047] |
| CD8 Tem-1         | T       | 26,714 | 40       | 667.9         | Head   | 0.187 [0.128, 0.250]     | 0.162 [0.107, 0.221] |
| CD14 Monocyte-1   | Myeloid | 29,871 | 40       | 746.8         | Head   | 0.437 [0.378, 0.495]     | 0.451 [0.392, 0.507] |
| CD56 NK           | NK      | 38,442 | 40       | 961.0         | Head   | 0.398 [0.320, 0.436]     | 0.437 [0.373, 0.480] |
| CD16 NK           | NK      | 58,507 | 40       | 1462.7        | Head   | 0.383 [0.326, 0.434]     | 0.376 [0.317, 0.429] |
| CD4 + Tcm-1       | T       | 70,653 | 40       | 1766.3        | Head   | 0.417 [0.368, 0.460]     | 0.435 [0.393, 0.473] |

For continuity with common long-tail reporting, Table 15 aggregates the seven least and seven most abundant subtypes. This summary is secondary to the per-class audit because individual subtypes can depart sharply from the abundance trend.

| Model                      | Tail Macro-F1 | Head Macro-F1 |
|----------------------------|---------------|---------------|
| Soft-cascade Transformer   | 0.142         | 0.284         |
| Flat FT-style Transformer  | 0.117         | 0.290         |
| FT-style Transformer + HCE | 0.125         | 0.288         |
| XGBoost                    | 0.119         | 0.292         |
| Logistic Regression        | 0.114         | 0.197         |
| Linear SVM                 | 0.082         | 0.204         |
| Random Forest              | 0.072         | 0.245         |
| Reference Correlation      | 0.060         | 0.114         |

Table 15: Fine-subtype macro-F1 on the seven least and seven most abundant subtypes under cell-weighted pooled patient-disjoint evaluation. This aggregate is reported as a descriptive complement to the complete per-subtype audit in Table 14.

**Cross-lineage error relative to an error-conditioned null.** Table 16 uses the wrong-conditioned null as the primary reference: after excluding the true subtype, it renormalizes the predicted-subtype marginal so every randomized outcome remains an error. For the four main models, observed cross-lineage rates are 0.743–0.757 versus wrong-conditioned expectations of 0.741–0.745. The corresponding differences are only +0.002 to +0.013, and all four patient-bootstrap 95% intervals include zero. Thus the raw approximately 75% rate is largely explained by the available wrong-class distribution and does not independently demonstrate either hierarchy preservation or hierarchy failure. The unconditional permutation, which can create chance exact matches and therefore uses a different sample space, is retained only as a sensitivity analysis.

| Model                      | Errors  | Observed | Wrong-cond. null | Obs. – WC null [95% CI] | Uncond. null (sens.) |
|----------------------------|---------|----------|------------------|-------------------------|----------------------|
| Soft-cascade Transformer   | 251,354 | 0.750    | 0.744            | +0.005 [-0.025, +0.028] | 0.675                |
| Flat FT-style Transformer  | 247,798 | 0.757    | 0.745            | +0.011 [-0.020, +0.035] | 0.675                |
| FT-style Transformer + HCE | 250,944 | 0.743    | 0.741            | +0.002 [-0.028, +0.023] | 0.671                |
| XGBoost                    | 244,990 | 0.754    | 0.741            | +0.013 [-0.020, +0.038] | 0.672                |
| Logistic Regression        | 301,566 | 0.705    | 0.752            | -0.047 [-0.061, -0.036] | 0.720                |
| Linear SVM                 | 296,904 | 0.735    | 0.774            | -0.039 [-0.053, -0.028] | 0.738                |
| Random Forest              | 246,420 | 0.809    | 0.749            | +0.059 [+0.021, +0.083] | 0.685                |
| Reference Correlation      | 328,596 | 0.775    | 0.785            | -0.009 [-0.022, +0.003] | 0.758                |

Table 16: Cross-lineage error relative to an error-conditioned marginal null. The primary wrong-conditioned (WC) null draws from each model’s predicted-subtype marginal after excluding the true subtype and renormalizing, so every randomized outcome remains wrong. Observed – WC-null confidence intervals recompute both terms in 2,000 patient-clustered bootstrap resamples. The unconditional permutation preserves the realized true- and predicted-subtype marginals but can create chance exact matches; it is reported only as a sensitivity analysis.

#### C.10 PROTOCOL SENSITIVITY: PATIENT-LEVEL VERSUS CELL-LEVEL SPLITTING

Table 17 compares the primary patient-disjoint protocol with a leakage-prone cell-level split control. It is placed in the appendix because the central result uses patient-disjoint evaluation throughout; the control quantifies how much the easier protocol would inflate both absolute performance and the apparent advantage of a model family.

**Metric definitions.** Let  $C^{(p)} \in \mathbb{R}^{K \times K}$  be the confusion matrix for held-out subject  $p$  and  $n_p = \sum_{ij} C_{ij}^{(p)}$ . We distinguish

$$C^{\text{CW}} = \sum_p C^{(p)}, \quad F_1^{\text{CW}} = \text{MacroF1}(C^{\text{CW}}), \quad (14)$$

$$C^{\text{SB}} = \frac{1}{P} \sum_p \frac{C^{(p)}}{n_p}, \quad F_1^{\text{SB}} = \text{MacroF1}(C^{\text{SB}}), \quad (15)$$

$$F_1^{\text{within}} = \frac{1}{P} \sum_p \text{MacroF1}(C^{(p)}). \quad (16)$$

We call these *cell-weighted pooled Macro-F1*, *subject-balanced pooled Macro-F1*, and *mean within-subject Macro-F1*. Subject balancing normalizes each subject confusion matrix to unit mass before pooling; it is neither equal-cell subsampling nor averaging per-subject Macro-F1. Released files retain the legacy column name `patient_balanced_macro_f1` for  $F_1^{\text{SB}}$ , but no different patient-balanced statistic is used.

| Task           | Model                    | Patient-disjoint | Cell split | Inflation |
|----------------|--------------------------|------------------|------------|-----------|
| Coarse lineage | XGBoost                  | 0.322            | 0.372      | +0.050    |
|                | Soft-cascade Transformer | 0.326            | 0.417      | +0.090    |
| Fine subtype   | XGBoost                  | 0.143            | 0.212      | +0.069    |
|                | Soft-cascade Transformer | 0.152            | 0.260      | +0.108    |

Table 17: Macro-F1 under the primary patient-disjoint protocol versus a leakage-prone cell-level split control. Cell-level splitting inflates performance for both methods, and does so more strongly for Soft-cascade Transformer than for XGBoost. This demonstrates that evaluation protocol can change both absolute performance and the apparent relative advantage of model families, which is why APT-Bench treats patient-disjoint evaluation as a requirement rather than a reporting option.

**Equal-cell evaluation sensitivity.** Patient-disjoint evaluation prevents leakage but does not make a cell-weighted pooled point estimate subject-balanced; patient cell counts range from 1,929 to 22,627. Table 18 therefore recomputes both hierarchy endpoints after sampling 1,900 cells without

replacement from every patient for each of 50 seeds. Coarse macro-F1 changes by at most +0.005, while fine macro-F1 changes by  $-0.007$  to  $-0.009$ ; the qualitative ordering in this sensitivity is unchanged. These small changes describe robustness of the historical cell-weighted scores to equal-cell sampling; they do not prove absence of patient influence. The main table now uses exact subject-normalized confusion pooling rather than this Monte Carlo subsampling approximation.

| Task           | Model                      | CW-pooled M-F1 | Equal-cell M-F1 [95% interval] | $\Delta$ |
|----------------|----------------------------|----------------|--------------------------------|----------|
| Coarse lineage | Soft-cascade Transformer   | 0.326          | 0.331 [0.325, 0.337]           | +0.005   |
|                | Flat FT-style Transformer  | 0.323          | 0.326 [0.322, 0.330]           | +0.003   |
|                | FT-style Transformer + HCE | 0.333          | 0.334 [0.328, 0.340]           | +0.000   |
|                | XGBoost                    | 0.322          | 0.324 [0.320, 0.329]           | +0.002   |
| Fine subtype   | Soft-cascade Transformer   | 0.152          | 0.144 [0.138, 0.153]           | -0.008   |
|                | Flat FT-style Transformer  | 0.147          | 0.138 [0.131, 0.147]           | -0.008   |
|                | FT-style Transformer + HCE | 0.151          | 0.142 [0.137, 0.151]           | -0.009   |
|                | XGBoost                    | 0.143          | 0.136 [0.131, 0.144]           | -0.007   |

Table 18: Equal-cell evaluation sensitivity. For each of 50 seeds, 1,900 cells are sampled without replacement from every subject, and cell-weighted pooled Macro-F1 is recomputed on the resulting 76,000 cells. All models use identical sampled row indices. The interval is the empirical 2.5th–97.5th percentile across seeds;  $\Delta$  is the equal-cell mean minus the original cell-weighted pooled metric. This resampling diagnostic is not the subject-balanced pooled metric, and it does not retrain or reselect models.

#### C.11 SUPPORTING SUBJECT-CELL SAMPLING AUDIT

The following analyses address sampling-policy sensitivity, not the clinical validity of disease prediction. They preserve the completed cross-cohort evidence as support for the cell-to-patient evaluation.

**Two allocation estimands.** For a specified sampling policy  $\pi$  and learning procedure  $f$ , define

$$\Delta_{\pi}(T) = \mathbb{E}[S(f_{\pi,32,T}) - S(f_{\pi,8,T})], \quad (17)$$

where  $S$  is subject-balanced pooled macro-F1 and the expectation averages training-subset randomization under the fixed cohort and evaluation protocol. Natural sampling estimates a *total allocation effect*, including label distribution changes. The matched policy estimates a *label-standardized allocation effect* relative to the sampled  $P = 8$  quota vector, averaged over these reference draws. It also imposes within-class donor round robin. These are different policy targets, not competing estimators of a uniquely true patient effect; neither isolates biological diversity from batch structure.

**Stability is not uncertainty of the mean.** The released joint stability interval is the central 95% range obtained by selecting one completed subset seed and bootstrapping paired test subjects. It retains between-subset variability rather than averaging resampled seeds. Its zero inclusion does not test  $\Delta_{\pi}(T) = 0$ , and its positive-replicate fraction is neither a posterior probability nor a significance test. Inference for the seed-averaged effect requires a separately constructed interval; the present main comparisons report stability, not that inferential claim.

#### C.12 FIXED TOTAL CELLS DO NOT ISOLATE SUBJECT BREADTH

We begin with the conventional label-agnostic design: outer-test subjects remain fixed while a constant number of training cells is reallocated from 8 to 32 subjects. Across APT, COMBAT, and OneK1K, 37 of 38 model–resolution–budget comparisons favor greater subject breadth. The direction appears strikingly consistent across assays and cohorts.

It is not an isolated subject effect. Increasing the number of sampled subjects also changes which labels are observed, how the fixed cell budget is distributed among classes, and the resulting class proportions. For example, the mean number of observed fine classes rises with subject count in all three datasets, despite fixed total cells. Training-subset intervals are also much less stable than intervals that condition on the averaged training subsets and bootstrap only test subjects. The label-agnostic

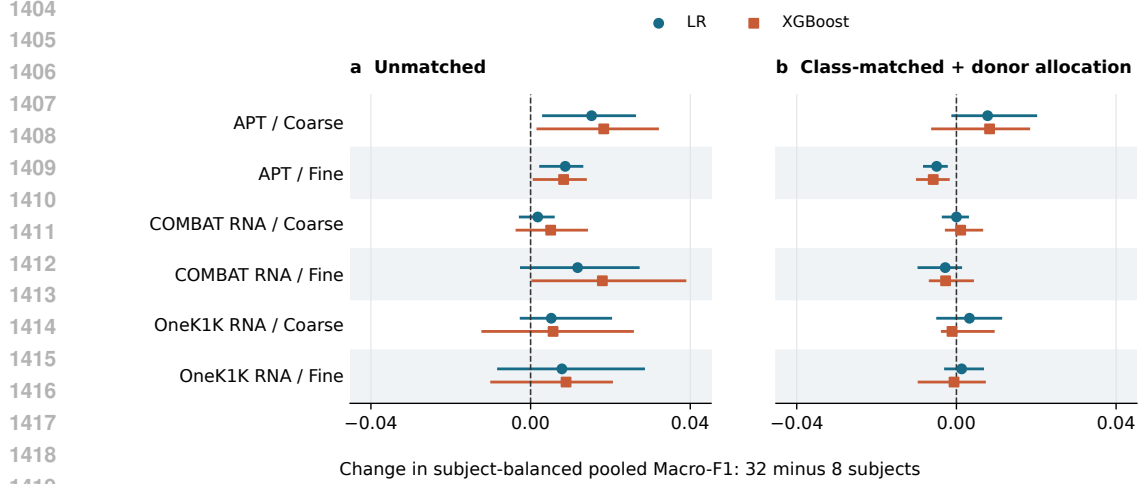


Figure 3: The apparent breadth advantage changes under the matched protocol. Both panels show the change in subject-balanced pooled OOF macro-F1 from 8 to 32 training subjects at  $T = 6,400$ , on identical axes. The right panel fixes class support and exact quotas and additionally uses within-class donor round robin; this is not a label-only intervention. Points are seed means; bars in both panels are empirical 2.5th–97.5th seed percentiles, not confidence intervals of the mean. Unmatched runs and matched LR use 20 seeds; matched XGBoost uses 10. APT and COMBAT fine point estimates change sign for both models. Cross-panel changes are descriptive, not paired attenuation tests. The matched joint stability intervals, which additionally resample held-out subjects, all span zero (Appendix Figure 5).

experiment therefore measures a total allocation effect, not a label-standardized effect. Conditioning on completed training subsets also omits sensitivity to alternative development samples.

### C.13 BREADTH ESTIMATES DEPEND ON THE ALLOCATION POLICY

The matched comparison freezes the  $P = 8$  observed class set and exact per-class quotas, then samples those same quotas from 8 and 32 subjects. Every joint replicate first selects one completed training-subset seed and then resamples paired outer-test subjects. This targets the residual effect of redistributing a fixed class-matched cell budget across more subjects. The resulting interval describes joint stability, not precision of the seed mean.

The matched policy yields a different pattern (Figure 3). At  $T = 6,400$ , LR/XGBoost effects are  $+0.0078/+0.0083$  for APT coarse but  $-0.0050/-0.0058$  for APT fine. COMBAT effects are approximately zero for coarse and  $-0.0028/-0.0027$  for fine. OneK1K effects are small and change sign across models. All 12 joint stability intervals include zero. The APT and COMBAT fine point-estimate reversals are therefore not specific to a linear classifier, but their signs are not stable across joint resampling. This is a joint sampling-protocol contrast: the implementation fixes observed labels, class counts, and proportions and also uses within-class donor round robin. The attenuation cannot therefore be assigned solely to label distribution. An independently controlled synthetic check separates quota matching from donor allocation (Appendix C.4); it does not replace this separation on the real cohorts. A complementary known-null validation yields mean effects  $+0.0202$ ,  $+0.0119$ , and  $+0.0004$  for unmatched, support-matched, and exact-quota sampling against an exactly zero overlap-quota target. Independent Monte Carlo references under donor shifts remain positive and are not mechanically removed by matching (Appendix C.5). This checks recovery of a specified allocation target, not interval coverage or a universal causal effect of subject breadth.

Beyond the common 32-subject grid, class-matched LR frontiers through 64 COMBAT and 128 OneK1K training donors yield adjacent mean effects between  $-0.00065$  and  $+0.00123$ , with every joint stability interval spanning zero. These ranges do not test mean equivalence or a plateau. A post-hoc  $\pm 0.01$  sensitivity band contains 11 of 12 adjacent frontier stability intervals, but only 4 of 12 main  $T = 6,400$  intervals and none of the four APT intervals (Appendix C.18). These

are pointwise interval-containment diagnostics, not equivalence tests or an optimal-allocation result. Complete unmatched grids, class-support audits, matched doublings, extended frontiers, and uncertainty decompositions appear in Appendix C.14– C.17.

#### C.14 FAIR FIXED-TOTAL AND MATCHED-SCALE ANALYSIS

The supporting scaling design uses  $P \in \{8, 16, 32\}$  and  $C \in \{100, 200, 400, 800, 1600\}$ , with 20 nested subset seeds for every outer fold. Because every retained patient has at least 1,929 cells, all finite caps are feasible without replacement. Disease-stratified patient orderings are nested across  $P$ , uniform cell permutations are nested across  $C$ , no subtype label enters sampling, and preprocessing is refit on each sampled training set. The fixed-total diagonals satisfy  $PC \in \{3200, 6400, 12800\}$  exactly.

| Model   | Task   | Total cells | $P = 8$              | $P = 16$             | $P = 32$             |
|---------|--------|-------------|----------------------|----------------------|----------------------|
| LR      | Coarse | 3,200       | 0.270 [0.259, 0.280] | 0.277 [0.270, 0.285] | 0.282 [0.279, 0.286] |
| LR      | Coarse | 6,400       | 0.277 [0.266, 0.290] | 0.286 [0.278, 0.293] | 0.292 [0.290, 0.295] |
| LR      | Coarse | 12,800      | 0.281 [0.270, 0.294] | 0.291 [0.284, 0.300] | 0.298 [0.296, 0.301] |
| LR      | Fine   | 3,200       | 0.078 [0.073, 0.084] | 0.082 [0.079, 0.087] | 0.085 [0.083, 0.089] |
| LR      | Fine   | 6,400       | 0.080 [0.077, 0.087] | 0.085 [0.081, 0.089] | 0.089 [0.087, 0.091] |
| LR      | Fine   | 12,800      | 0.081 [0.076, 0.087] | 0.087 [0.083, 0.090] | 0.091 [0.089, 0.094] |
| XGBoost | Coarse | 3,200       | 0.262 [0.250, 0.276] | 0.272 [0.260, 0.284] | 0.279 [0.274, 0.284] |
| XGBoost | Coarse | 6,400       | 0.269 [0.258, 0.284] | 0.279 [0.267, 0.289] | 0.288 [0.285, 0.292] |
| XGBoost | Coarse | 12,800      | 0.276 [0.267, 0.290] | 0.289 [0.277, 0.298] | 0.298 [0.296, 0.300] |
| XGBoost | Fine   | 3,200       | 0.082 [0.078, 0.088] | 0.087 [0.081, 0.092] | 0.089 [0.086, 0.092] |
| XGBoost | Fine   | 6,400       | 0.086 [0.081, 0.094] | 0.091 [0.086, 0.097] | 0.095 [0.090, 0.098] |
| XGBoost | Fine   | 12,800      | 0.092 [0.087, 0.098] | 0.097 [0.092, 0.102] | 0.098 [0.096, 0.101] |

Table 19: Fixed-total training-cell scaling. Each row holds the total number of sampled training cells constant while reallocating them across independent patients. Entries are subject-balanced pooled, patient-disjoint OOF Macro-F1 means with empirical 95% intervals over 20 matched subset seeds. For each score, every held-out subject’s confusion matrix is normalized to unit mass before pooling; this is distinct from both cell-weighted pooling and mean within-subject Macro-F1. All outer-test cells are evaluated.

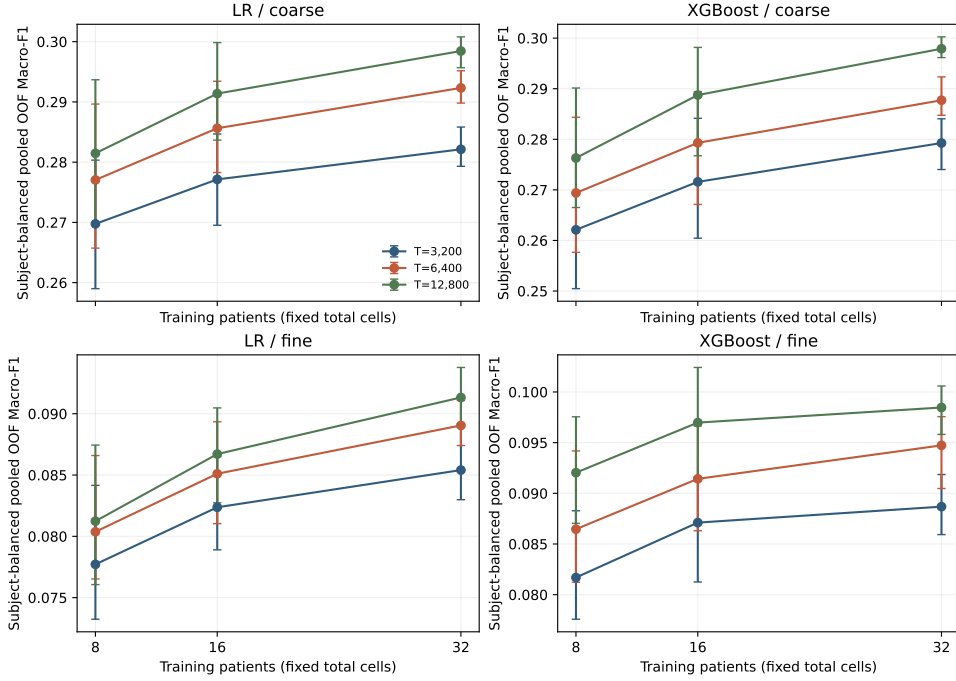


Figure 4: APT-only fixed-total response curves underlying the unmatched cross-cohort diagnostic in Table 20. Points are pooled OOF macro-F1 means over 20 matched subset seeds; bars are empirical 95% seed intervals. Patient-balanced and patient-bootstrap results are provided in the machine-readable artifacts.

Table 19 reports variability across subset seeds. Separately, paired patient-clustered bootstraps over the same 40 pooled OOF patients compare  $P = 8$  with  $P = 32$  after averaging seed-level sufficient statistics. All 12 intervals exclude zero; their lower endpoints range from 0.0002 to 0.0084 and upper endpoints from 0.0090 to 0.0314. These intervals quantify uncertainty over patients, whereas the empirical seed intervals quantify sensitivity to which development patients and cells are sampled.

Matched doubling effects average patient versus cell gains of 0.011/0.005 (LR coarse), 0.007/0.003 (LR fine), 0.016/0.008 (XGBoost coarse), and 0.009/0.006 (XGBoost fine). In the descriptive fold-fixed-effect response surface, the corresponding per-doubling  $\beta_P/\beta_C$  coefficients are 0.017/0.008, 0.008/0.004, 0.016/0.007, and 0.009/0.006. Complete pairwise effects, patient intervals, seed intervals, and response-surface uncertainty are stored in the machine-readable artifacts. The coefficients are associations over this grid, not causal or universal scaling laws.

#### C.15 EXTERNAL COMBAT AND ONEK1K SCALING REPLICATIONS

We downloaded the author-released COMBAT CITE-seq object (COVID-19 Multi-omics Blood Atlas (COMBAT) Consortium, 2022) (836,148 cells and 124 donor identifiers; SHA-256 f628a15f25b9ca2f7cdefeab271fd9d007c8d5e47eb80c4b807d5f65e86ff53d). We removed cells without a resolved major or fine immune annotation and cells flagged as GEX-region doublets. Ten donors were frozen as a calibration-only set and excluded from every model fit and test fold. Among the remaining donors, requiring at least 1,600 eligible cells leaves 105 donors and 700,706 cells. We construct five disjoint folds of 21 test donors, stratified by the released sample-source field, and freeze these folds across all models, budgets, seeds, and modalities.

The RNA replication uses 293 protein-coding genes with highest variance on the 10 calibration donors after excluding mitochondrial and ribosomal genes. This matches the APT input dimension without selecting features on any evaluation donor. The ADT sensitivity uses all 192 released antibody features. Both inputs use the released five-class major annotation and 39-class fine an-

notation. They share the exact  $P$ ,  $C$ , and fixed-total grids, 20 nested source-stratified donor/cell seeds, training-only standardization, LR configuration, and paired donor-bootstrap procedure used in APT-Bench. The RNA analysis additionally uses the frozen 400-tree XGBoost baseline. Every fit is evaluated on all cells of the 21 untouched test donors; the ADT analysis is an LR-only modality sensitivity rather than an additional model-comparison family. Dataset hashes, folds, features, sampling manifests, and per-donor sufficient statistics are included in the release artifacts.

OneK1K provides a second independent scRNA-seq replication in a much larger healthy PBMC cohort (Yazar et al., 2022). We use the current author-deposited CELLxGENE object (1,248,980 cells and 981 donor identifiers; SHA-256 14c5e9e70007a5e7dc047f7b29405773305e1a7b9791d0789af70649cf6f46ec), remove released doublet calls, and reserve one complete multiplexing pool for calibration. Requiring at least 800 eligible cells leaves 925 evaluation donors and 1,199,468 cells. Five folds are disjoint in both donor and multiplexing pool. We select 293 protein-coding, non-mitochondrial, non-ribosomal genes by variance on ten calibration-pool donors only, and use the released 30-class Azimuth-derived fine labels with a deterministic five-lineage mapping.

OneK1K uses the same models, 20 nested subset seeds, training-only standardization, untouched-test-donor evaluation, subject-balanced pooled metric, and 2,000 paired donor bootstraps. Its grid is  $P \in \{8, 16, 32\}$  and  $C \in \{100, 200, 400, 800\}$ , yielding the two common exact budgets  $T \in \{3200, 6400\}$ . We prespecified the lower depth ceiling because only 116 of the 981 raw donors have at least 1,600 eligible cells; enforcing the full APT grid would discard most donors and condition the replication on high cell recovery. OneK1K tests scaling portability in a healthy cohort, not disease generalization or absolute accuracy across different feature spaces and label-generation procedures.

| Dataset | Modality | Model/task     | $T = 3,200$             | $T = 6,400$             | $T = 12,800$            |
|---------|----------|----------------|-------------------------|-------------------------|-------------------------|
| APT     | Aptamer  | LR/Coarse      | +0.012 [+0.005, +0.021] | +0.015 [+0.007, +0.024] | +0.017 [+0.009, +0.026] |
| APT     | Aptamer  | LR/Fine        | +0.008 [+0.004, +0.011] | +0.009 [+0.005, +0.012] | +0.010 [+0.005, +0.013] |
| APT     | Aptamer  | XGBoost/Coarse | +0.017 [+0.008, +0.027] | +0.018 [+0.007, +0.030] | +0.022 [+0.009, +0.035] |
| APT     | Aptamer  | XGBoost/Fine   | +0.007 [+0.002, +0.011] | +0.008 [+0.002, +0.013] | +0.006 [+0.000, +0.011] |
| COMBAT  | RNA      | LR/Coarse      | +0.003 [+0.002, +0.004] | +0.002 [+0.001, +0.003] | +0.001 [+0.001, +0.002] |
| COMBAT  | RNA      | LR/Fine        | +0.012 [+0.009, +0.013] | +0.012 [+0.010, +0.012] | +0.007 [+0.005, +0.009] |
| COMBAT  | RNA      | XGBoost/Coarse | +0.006 [+0.005, +0.007] | +0.005 [+0.004, +0.006] | +0.004 [+0.003, +0.005] |
| COMBAT  | RNA      | XGBoost/Fine   | +0.016 [+0.013, +0.017] | +0.018 [+0.016, +0.019] | +0.017 [+0.014, +0.018] |
| COMBAT  | ADT      | LR/Coarse      | +0.006 [+0.004, +0.008] | +0.009 [+0.006, +0.012] | +0.012 [+0.010, +0.016] |
| COMBAT  | ADT      | LR/Fine        | +0.013 [+0.010, +0.014] | +0.011 [+0.010, +0.013] | +0.011 [+0.009, +0.013] |
| OneK1K  | RNA      | LR/Coarse      | +0.004 [+0.004, +0.005] | +0.005 [+0.005, +0.006] | –                       |
| OneK1K  | RNA      | LR/Fine        | +0.007 [+0.007, +0.009] | +0.008 [+0.007, +0.009] | –                       |
| OneK1K  | RNA      | XGBoost/Coarse | -0.003 [-0.004, -0.002] | +0.006 [+0.005, +0.006] | –                       |
| OneK1K  | RNA      | XGBoost/Fine   | +0.006 [+0.005, +0.007] | +0.009 [+0.007, +0.009] | –                       |

Table 20: Unmatched diagnostic: change in subject-balanced pooled, subject-disjoint OOF Macro-F1 when the same total training-cell budget is reallocated from 8 to 32 independent subjects without freezing the marginal training label distribution. Each held-out subject confusion matrix is normalized to unit mass before pooling. Brackets are 95% paired subject-bootstrap intervals over the fixed outer-test subjects. Positive values favor broader subject coverage but can also reflect changes in observed classes, per-class counts, and class proportions. COMBAT ADT is a prespecified modality sensitivity using LR only. OneK1K uses the two common exact budgets supported by at least 800 cells from nearly all donors.

## C.16 LABEL-DISTRIBUTION AUDIT AND MATCHED CONTROL

The label-agnostic grid samples cells uniformly within subjects and therefore allows the full marginal label distribution to vary. Machine-readable audit tables report the observed class count, unseen-at-training class count, cell count, and donor count for every class at every fold, seed, and scaling point. The main text reports the common fixed-total diagonals, where fine-class coverage increases slightly with  $P$  despite fixed  $T$ ; this coverage statistic is only one observable component of the broader label-distribution effect.

The class-matched control fixes the marginal label distribution. The  $P = 8$  subset defines the observed labels and a proportional integer quota  $T_k$  with a one-cell floor; capacity violations are deterministically redistributed. The same quota vector is then used at  $P = 8, 16, 32$ . Within each

class, a capacity-constrained round-robin sampler distributes cells across selected subjects before assigning a second cell from the same subject, so increasing  $P$  changes realized donor coverage rather than class counts. Sampling is without replacement, standardization is refit on each training subset, and all immutable outer-test cells are retained. All three runs complete 1,200/1,200 jobs and pass checks for exact total cells, identical paired class counts, non-overlapping test subjects, and contribution by every nominal training subject. These primary Logistic Regression runs use 20 matched subset seeds at both  $T = 3,200$  and  $T = 6,400$ . To test whether the matched-policy direction is specific to a linear classifier, we repeat the identical control with the frozen XGBoost configuration at  $T = 6,400$  using 10 matched seeds. Each dataset completes 300/300 additional jobs and passes the same checks; final production fits use the CPU histogram backend throughout to avoid mixing computational backends. It does not separately identify changes in disease-subtype composition, within-class allocation across subjects, or within-class subject heterogeneity; these remain bundled with the subject-breadth intervention.

| Dataset    | Model   | Task   | $T$   | Classes (mean/total) | Donors/class ( $P$ : 8→32) | $\Delta$ SB-pooled M-F1    | $\Pr(\Delta > 0)$ |
|------------|---------|--------|-------|----------------------|----------------------------|----------------------------|-------------------|
| APT        | LR      | Coarse | 3,200 | 5.0/5                | 7.9→28.5                   | +0.0062 [-0.0080, +0.0221] | 0.778             |
| APT        | LR      | Coarse | 6,400 | 5.0/5                | 7.9→30.5                   | +0.0078 [-0.0070, +0.0265] | 0.836             |
| APT        | LR      | Fine   | 3,200 | 26.8/27              | 7.1→25.0                   | -0.0024 [-0.0077, +0.0029] | 0.163             |
| APT        | LR      | Fine   | 6,400 | 26.8/27              | 7.2→27.4                   | -0.0050 [-0.0105, +0.0009] | 0.043             |
| APT        | XGBoost | Coarse | 6,400 | 5.0/5                | 7.9→30.5                   | +0.0083 [-0.0130, +0.0265] | 0.817             |
| APT        | XGBoost | Fine   | 6,400 | 26.8/27              | 7.2→27.3                   | -0.0058 [-0.0143, +0.0027] | 0.110             |
| COMBAT RNA | LR      | Coarse | 3,200 | 5.0/5                | 8.0→29.0                   | +0.0011 [-0.0036, +0.0084] | 0.611             |
| COMBAT RNA | LR      | Coarse | 6,400 | 5.0/5                | 8.0→31.3                   | +0.0000 [-0.0055, +0.0045] | 0.547             |
| COMBAT RNA | LR      | Fine   | 3,200 | 39.0/39              | 7.2→22.5                   | -0.0007 [-0.0088, +0.0092] | 0.439             |
| COMBAT RNA | LR      | Fine   | 6,400 | 39.0/39              | 7.3→26.6                   | -0.0028 [-0.0114, +0.0043] | 0.272             |
| COMBAT RNA | XGBoost | Coarse | 6,400 | 5.0/5                | 8.0→31.2                   | +0.0011 [-0.0054, +0.0088] | 0.522             |
| COMBAT RNA | XGBoost | Fine   | 6,400 | 39.0/39              | 7.3→26.6                   | -0.0027 [-0.0096, +0.0066] | 0.264             |
| OneK1K RNA | LR      | Coarse | 3,200 | 5.0/5                | 7.8→27.7                   | +0.0025 [-0.0079, +0.0120] | 0.669             |
| OneK1K RNA | LR      | Coarse | 6,400 | 5.0/5                | 7.8→29.6                   | +0.0033 [-0.0059, +0.0128] | 0.728             |
| OneK1K RNA | LR      | Fine   | 3,200 | 29.0/30              | 6.2→19.8                   | +0.0019 [-0.0080, +0.0105] | 0.700             |
| OneK1K RNA | LR      | Fine   | 6,400 | 29.0/30              | 6.4→22.1                   | +0.0013 [-0.0058, +0.0089] | 0.598             |
| OneK1K RNA | XGBoost | Coarse | 6,400 | 5.0/5                | 7.8→29.5                   | -0.0011 [-0.0061, +0.0139] | 0.209             |
| OneK1K RNA | XGBoost | Fine   | 6,400 | 28.9/30              | 6.4→22.0                   | -0.0006 [-0.0117, +0.0098] | 0.549             |

Table 21: Class-matched fixed-total control. The outcome is subject-balanced pooled Macro-F1, computed by normalizing each held-out subject confusion matrix to unit mass before pooling. LR uses 20 seeds at both totals; the matched XGBoost robustness check uses 10 seeds at  $T = 6,400$ . Within each outer-fold/seed/task pair, the  $P = 8$  subset freezes the observed classes and exact per-class cell quotas; the same quotas are used at  $P = 16$  and  $P = 32$ . Brackets are joint 95% stability intervals obtained by resampling a training-subset seed and then paired outer-test subjects; the final column is the empirical fraction of joint replicates with a positive effect. These ranges are not confidence intervals for seed-averaged effects. Seed-only and subject-only intervals are released separately. Donors/class reports the mean number of subjects contributing cells to an observed class.

The primary stability diagnostic combines both variation sources. In each of 2,000 replicates, we sample one completed training-subset seed uniformly, resample outer-test subjects with replacement, use the same subject draw for  $P = 8$  and  $P = 32$ , and recompute the paired difference. All 18 joint 95% stability intervals include zero, and the table reports the empirical positive-replicate fraction. For decomposition, empirical seed-only intervals include zero in 16 of 18 rows; the exceptions are the two negative APT fine results at  $T = 6,400$ . Subject-only intervals condition on the seed-averaged prediction matrix and can exclude zero even when the joint stability interval does not. The constructions answer different questions: the joint range retains variation between fitted training subsets rather than estimating uncertainty of their average. Neither zero inclusion nor the positive-replicate fraction is a test of the mean effect. These ranges also lack a demonstrated coverage guarantee for a future training cohort; they are conditional resampling diagnostics.

#### C.17 EXTENDED EXTERNAL SUBJECT FRONTIER

The common  $P \leq 32$  grid is imposed by the 40-patient APT cohort, not by the capacity of the external datasets. We therefore extend the class-matched LR control to  $P = 64$  for COMBAT and to  $P \in \{64, 128\}$  for OneK1K at  $T \in \{3,200, 6,400\}$ . The original  $P = 8$  subset freezes the class set and integer per-class quotas for every fold, seed, task, and total; only the number of selected subjects allowed to contribute those cells changes. Each new point uses 10 nested subset seeds,

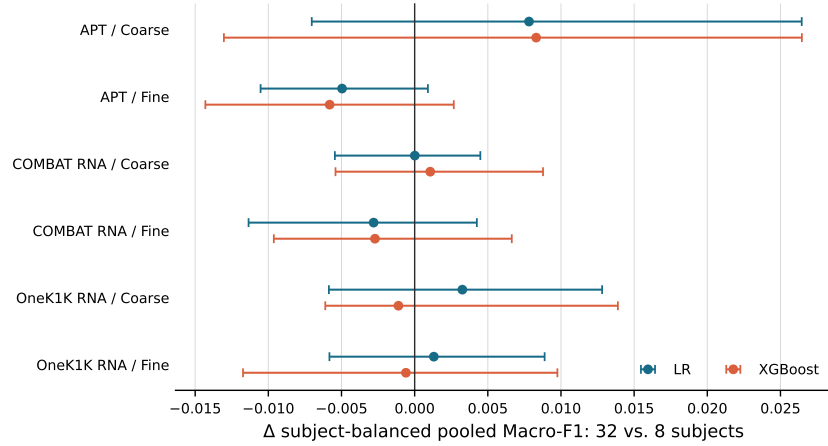


Figure 5: Joint stability for the completed class-matched protocol at  $T = 6,400$ . Points are mean changes in subject-balanced pooled OOF macro-F1 from 8 to 32 subjects. The 95% joint stability intervals select one training-subset seed and bootstrap paired held-out subjects; all 12 span zero. Unlike the seed-only bars in Figure 3, these intervals include test-subject variability. They are not confidence intervals for the seed-averaged effect; zero inclusion does not test a zero mean or establish equivalence.

all immutable outer-test subjects, and the same joint resampling of training-subset seed and held-out subjects used in the primary matched control. This extension is an external-cohort allocation sensitivity rather than a claim that its largest tested  $P$  is globally optimal.

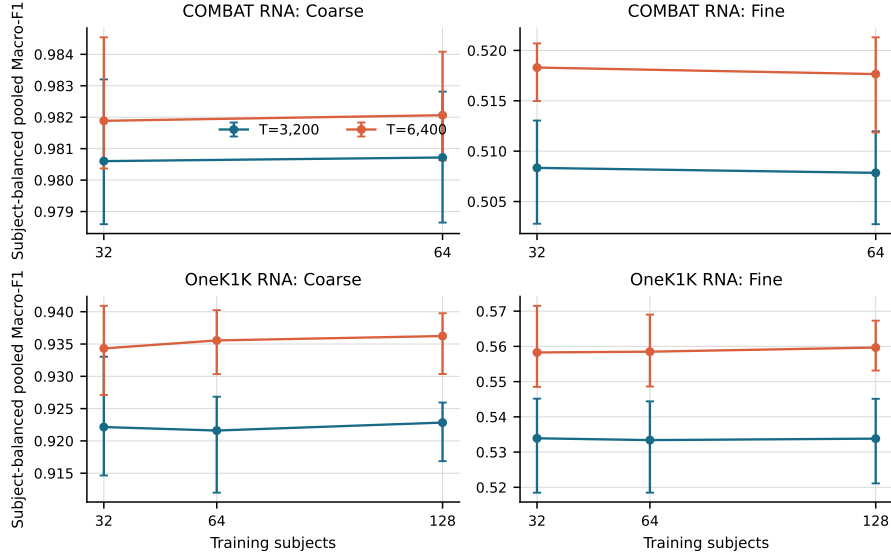


Figure 6: Class-matched LR frontier beyond 32 training subjects. Points are mean subject-balanced pooled OOF macro-F1 over 10 matched training-subset seeds; bars are empirical 95% seed intervals and therefore visualize subset sensitivity, not uncertainty of the mean. The paired joint stability intervals are reported in Table 22.

| Dataset    | Task   | $T$   | Subjects | $\Delta$ SB-pooled M-F1    | $\Pr(\Delta > 0)$ |
|------------|--------|-------|----------|----------------------------|-------------------|
| COMBAT RNA | Coarse | 3,200 | 32→64    | +0.0001 [-0.0020, +0.0025] | 0.535             |
| COMBAT RNA | Coarse | 6,400 | 32→64    | +0.0002 [-0.0014, +0.0016] | 0.618             |
| COMBAT RNA | Fine   | 3,200 | 32→64    | -0.0005 [-0.0060, +0.0026] | 0.498             |
| COMBAT RNA | Fine   | 6,400 | 32→64    | -0.0006 [-0.0060, +0.0040] | 0.450             |
| OneK1K RNA | Coarse | 3,200 | 32→64    | -0.0005 [-0.0111, +0.0052] | 0.553             |
| OneK1K RNA | Coarse | 6,400 | 32→64    | +0.0012 [-0.0046, +0.0046] | 0.766             |
| OneK1K RNA | Fine   | 3,200 | 32→64    | -0.0005 [-0.0060, +0.0053] | 0.418             |
| OneK1K RNA | Fine   | 6,400 | 32→64    | +0.0002 [-0.0096, +0.0063] | 0.595             |
| OneK1K RNA | Coarse | 3,200 | 64→128   | +0.0012 [-0.0023, +0.0068] | 0.642             |
| OneK1K RNA | Coarse | 6,400 | 64→128   | +0.0007 [-0.0042, +0.0043] | 0.657             |
| OneK1K RNA | Fine   | 3,200 | 64→128   | +0.0004 [-0.0039, +0.0050] | 0.542             |
| OneK1K RNA | Fine   | 6,400 | 64→128   | +0.0012 [-0.0066, +0.0068] | 0.617             |

Table 22: Extended class-matched subject frontier for Logistic Regression. Each comparison holds the total training-cell budget, observed class set, and exact per-class cell quotas fixed while increasing the number of contributing training subjects. Values are paired mean changes; brackets are joint 95% stability intervals from 2,000 replicates that sample a matched training-subset seed and then bootstrap held-out subjects. The final column is the empirical fraction of positive replicates, not a posterior probability. These ranges are not confidence intervals for seed-averaged effects.

All 12 adjacent frontier effects are small: their means range from  $-0.00065$  to  $+0.00123$ , and every joint stability interval includes zero. The four direct OneK1K  $32 \rightarrow 128$  contrasts likewise span zero under joint resampling. This does not establish a plateau or exclude practically meaningful changes. The explicit margin sensitivity below assesses interval width separately from zero inclusion. Because the intervention remains tied to these tasks, budgets, features, and LR configuration, it does not locate a universal saturation point or optimal allocation.

#### C.18 POST-HOC STABILITY-BAND SENSITIVITY

Failure to detect a difference is not evidence of equivalence (Lakens, 2017). After inspecting the existing results, we add an *exploratory* smallest-effect sensitivity bound  $\epsilon = 0.01$  absolute subject-balanced pooled Macro-F1 (one percentage point), with 0.005 and 0.02 as sensitivity values. This is a reporting tolerance, not a validated clinical threshold, and was not prespecified for the completed experiments. It must not be conflated with the separate compact-panel tolerance  $\delta = 0.02$ .

For each released joint 95% stability interval  $[L, U]$ , we report zero inclusion ( $L \leq 0 \leq U$ ) and strict containment ( $-\epsilon < L$  and  $U < \epsilon$ ) separately. Containment describes concentration of this conditional resampling distribution inside the chosen band, not equivalence of the mean allocation effects. Intervals with  $L \leq -\epsilon$  and  $U \geq \epsilon$  span changes of the stated magnitude in both directions. These diagnostics neither quantify statistical power nor provide a calibrated future-cohort guarantee.

At  $T = 6,400$ , only 4 of 12 main comparison intervals meet the  $\pm 0.01$  containment criterion; none of the four APT intervals does. The APT coarse XGBoost interval spans both meaningful directions. Among adjacent external LR frontier comparisons, 11 of 12 intervals meet this criterion; the exception is OneK1K coarse  $32 \rightarrow 64$  at  $T = 3,200$ . With the tighter  $\pm 0.005$  margin, no main interval and only 4 of 12 adjacent frontier intervals are contained. Descriptive stability is therefore margin-dependent; these counts do not establish that the sampling policies are equivalent or subjects have no value.

We do not report formal TOST  $p$ -values. Conventional level-0.05 TOST is linked to a corresponding 90% confidence interval, whereas this analysis retains the released joint 95% stability intervals. More fundamentally, those intervals sample one completed training-subset seed and then test subjects, combining between-fit variability with test uncertainty rather than estimating only a seed-averaged mean effect. Containment is therefore an exploratory joint-interval screen, not a calibrated confirmatory equivalence test. Results are pointwise, without a simultaneous equivalence guarantee across rows, and do not establish a continuous plateau between or beyond the tested budgets. This audit concerns the completed legacy controls, not the ongoing A–D GPU rerun.

| Comparison  | Dataset | Task   | $n$ | $\pm .005$ | $\pm .010$ | $\pm .020$ | Both signs* |
|-------------|---------|--------|-----|------------|------------|------------|-------------|
| 8-32        | APT     | coarse | 2   | 0          | 0          | 0          | 1           |
| 8-32        | APT     | fine   | 2   | 0          | 0          | 2          | 0           |
| 8-32        | COMBAT  | coarse | 2   | 0          | 2          | 2          | 0           |
| 8-32        | COMBAT  | fine   | 2   | 0          | 1          | 2          | 0           |
| 8-32        | OneK1K  | coarse | 2   | 0          | 0          | 2          | 0           |
| 8-32        | OneK1K  | fine   | 2   | 0          | 1          | 2          | 0           |
| Extended LR | COMBAT  | coarse | 2   | 2          | 2          | 2          | 0           |
| Extended LR | COMBAT  | fine   | 2   | 0          | 2          | 2          | 0           |
| Extended LR | OneK1K  | coarse | 4   | 2          | 3          | 4          | 0           |
| Extended LR | OneK1K  | fine   | 4   | 0          | 4          | 4          | 0           |

Table 23: Post-hoc stability-band sensitivity: number of released joint 95% stability intervals strictly inside each stated margin. The 8-32 rows use  $T = 6,400$  and both LR and XGBoost; extended rows use LR at both totals for adjacent subject-budget comparisons. \*Intervals reaching both  $-0.01$  and  $+0.01$ , indicating that the resampling range spans changes of this magnitude in both directions. Counts are pointwise diagnostics, not mean-effect confidence intervals, formal TOST decisions, or evidence of equivalence.

| Model                         | Task   | $C = 100$            | $C = 500$            | $C = 2000$           | $C = \text{all}$     |
|-------------------------------|--------|----------------------|----------------------|----------------------|----------------------|
| Logistic Regression, $P = 8$  | Coarse | 0.255 [0.247, 0.265] | 0.274 [0.263, 0.284] | 0.283 [0.269, 0.297] | 0.282 [0.269, 0.293] |
| Logistic Regression, $P = 16$ | Coarse | 0.266 [0.261, 0.270] | 0.290 [0.284, 0.297] | 0.288 [0.282, 0.295] | 0.290 [0.280, 0.300] |
| Logistic Regression, $P = 24$ | Coarse | 0.276 [0.269, 0.280] | 0.296 [0.288, 0.304] | 0.288 [0.282, 0.295] | 0.296 [0.289, 0.302] |
| Logistic Regression, $P = 32$ | Coarse | 0.283 [0.278, 0.288] | 0.299 [0.296, 0.302] | 0.287 [0.284, 0.288] | 0.299 [0.284, 0.315] |
| Logistic Regression, $P = 8$  | Fine   | 0.074 [0.068, 0.081] | 0.084 [0.079, 0.088] | 0.087 [0.082, 0.092] | 0.101 [0.091, 0.109] |
| Logistic Regression, $P = 16$ | Fine   | 0.081 [0.078, 0.085] | 0.090 [0.088, 0.094] | 0.095 [0.090, 0.099] | 0.116 [0.110, 0.125] |
| Logistic Regression, $P = 24$ | Fine   | 0.086 [0.083, 0.089] | 0.093 [0.089, 0.097] | 0.100 [0.095, 0.105] | 0.125 [0.118, 0.131] |
| Logistic Regression, $P = 32$ | Fine   | 0.089 [0.086, 0.092] | 0.095 [0.093, 0.097] | 0.105 [0.102, 0.108] | 0.132 [0.112, 0.146] |
| XGBoost, $P = 8$              | Coarse | 0.246 [0.234, 0.263] | 0.262 [0.250, 0.278] | 0.277 [0.262, 0.293] | 0.300 [0.283, 0.321] |
| XGBoost, $P = 16$             | Coarse | 0.260 [0.252, 0.272] | 0.278 [0.268, 0.288] | 0.295 [0.283, 0.306] | 0.313 [0.298, 0.330] |
| XGBoost, $P = 24$             | Coarse | 0.270 [0.264, 0.276] | 0.289 [0.284, 0.295] | 0.305 [0.297, 0.312] | 0.319 [0.311, 0.328] |
| XGBoost, $P = 32$             | Coarse | 0.275 [0.270, 0.280] | 0.296 [0.293, 0.298] | 0.310 [0.308, 0.312] | 0.322 [0.300, 0.345] |
| XGBoost, $P = 8$              | Fine   | 0.073 [0.068, 0.079] | 0.085 [0.080, 0.091] | 0.096 [0.090, 0.103] | 0.118 [0.110, 0.126] |
| XGBoost, $P = 16$             | Fine   | 0.082 [0.077, 0.085] | 0.096 [0.092, 0.101] | 0.110 [0.103, 0.120] | 0.132 [0.125, 0.140] |
| XGBoost, $P = 24$             | Fine   | 0.087 [0.081, 0.092] | 0.100 [0.095, 0.104] | 0.115 [0.110, 0.121] | 0.138 [0.132, 0.144] |
| XGBoost, $P = 32$             | Fine   | 0.090 [0.087, 0.093] | 0.102 [0.100, 0.105] | 0.118 [0.117, 0.121] | 0.143 [0.127, 0.159] |

Table 24: Patient-cell training scaling. Entries are pooled OOF Macro-F1 means with empirical 2.5th-97.5th seed intervals; the deterministic  $P = 32, C = \text{all}$  endpoint instead uses a patient-clustered bootstrap interval. Seed intervals are not patient confidence intervals.

### C.19 SECONDARY DESCRIPTIVE PATIENT-CELL SCALING SURFACE

This historical  $4 \times 4$  surface fixes eight outer-test patients and varies only the training set. For each outer fold and each of 20 seeds, we generate one approximately proportional disease-stratified ordering of the 32 development patients, giving nested sets  $S_8 \subset S_{16} \subset S_{24} \subset S_{32}$ . Each selected patient receives one uniform random cell ordering, giving nested prefixes of 100, 500, and 2,000 cells and the complete available set. Patients with fewer than a requested cap contribute all available cells. No cell is selected using its subtype, and every model is evaluated on all cells from the untouched test patients. The full-data endpoint is fitted once per fold rather than duplicated across seeds. All 6,020 model-task jobs completed, and the deterministic endpoint exactly reproduces the frozen pooled OOF baselines within absolute tolerance  $5 \times 10^{-4}$ . Because  $P=8 \rightarrow 32$  is a fourfold change whereas  $C=100 \rightarrow \text{all}$  is a larger and patient-dependent change, this surface is descriptive only. All direct comparisons between the two axes use the fixed-total and matched-doubling analysis in the main paper.

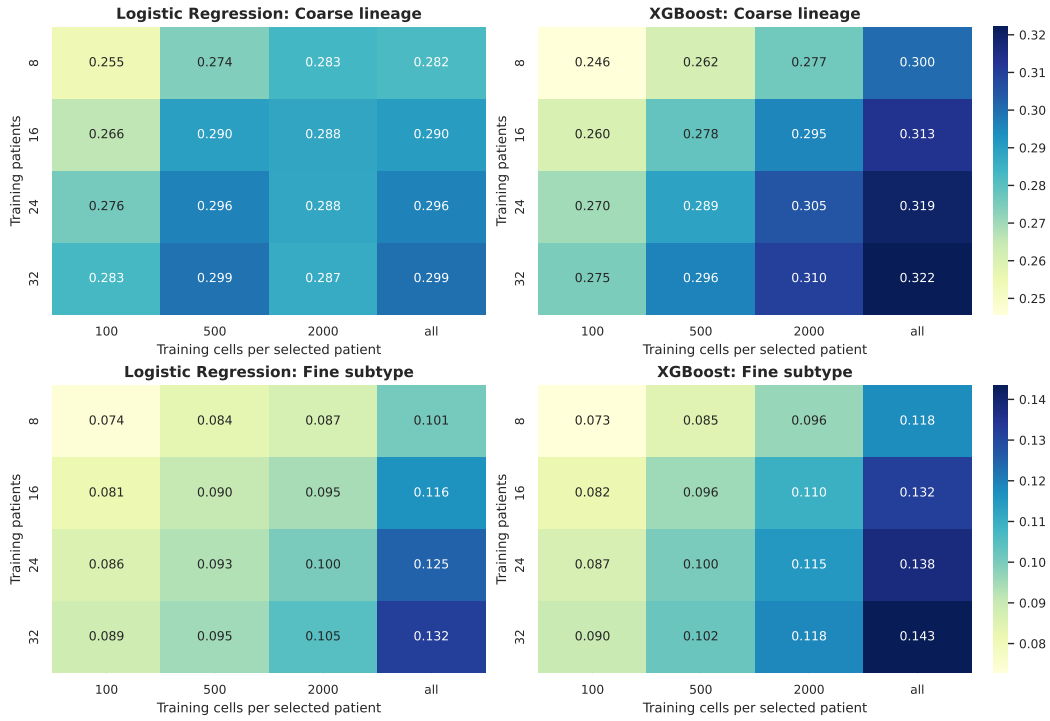


Figure 7: Cell-weighted pooled patient-disjoint Macro-F1 over the complete training-patient and per-patient training-cell grid. Values average 20 nested subset seeds except for the deterministic full-data endpoint. Color scales are shared across models within each task.

| Model                    | Input | Coarse SB-Macro-F1 $\uparrow$ | Fine SB-Macro-F1 $\uparrow$ |
|--------------------------|-------|-------------------------------|-----------------------------|
| Soft-cascade Transformer | APT   | 0.332 [0.306, 0.354]          | 0.145 [0.122, 0.160]        |
| Flat Transformer         | APT   | 0.326 [0.300, 0.348]          | 0.139 [0.114, 0.155]        |
| HCE Transformer          | APT   | 0.335 [0.310, 0.356]          | 0.143 [0.121, 0.160]        |
| XGBoost                  | APT   | 0.325 [0.302, 0.345]          | 0.137 [0.120, 0.149]        |
| Linear SVM               | APT   | 0.309 [0.292, 0.324]          | 0.105 [0.089, 0.118]        |
| Logistic Regression      | APT   | 0.298 [0.281, 0.315]          | 0.126 [0.107, 0.139]        |
| Random Forest            | APT   | 0.253 [0.227, 0.277]          | 0.089 [0.079, 0.097]        |
| Reference Correlation    | APT   | 0.171 [0.147, 0.192]          | 0.068 [0.050, 0.078]        |

Table 25: Complete subject-balanced pooled Macro-F1 reference inventory, including auxiliary classical baselines. Definitions, resampling, and scope match Table 1.

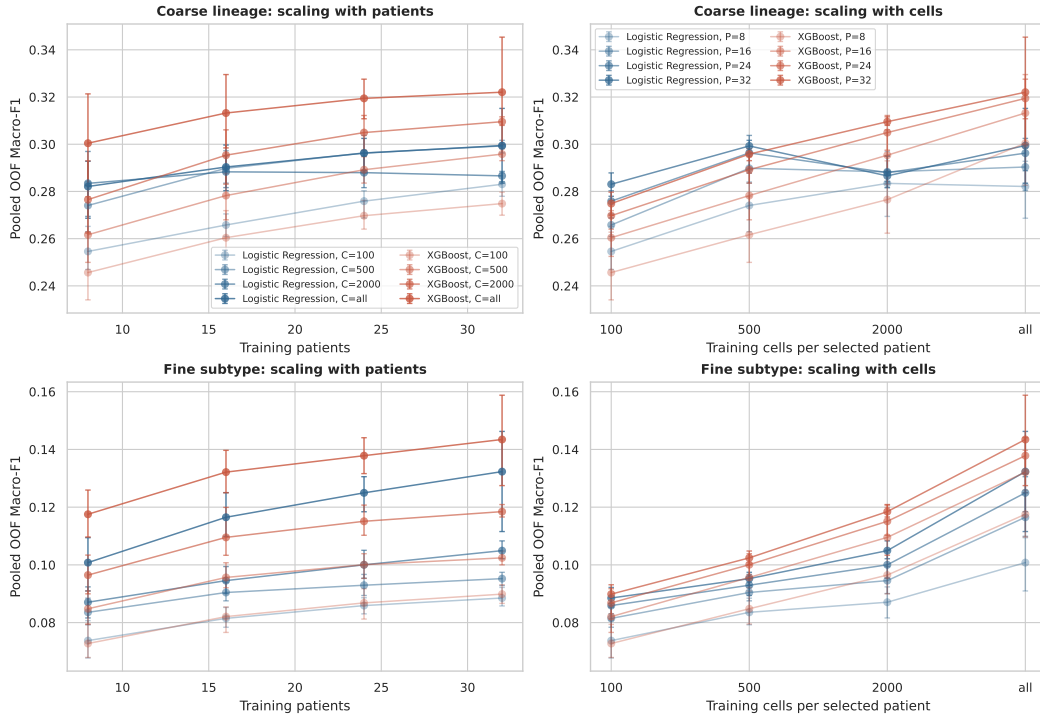


Figure 8: Patient-budget curves at each fixed per-patient cell cap. Intervals show subset-resampling variability; they do not represent uncertainty over a new patient population. Non-monotonic local changes are retained rather than smoothed.

The surface is cohort- and range-specific. We do not compare its unequal endpoints or use it to establish a patient- or cell-dominant scaling law, and it cannot estimate the benefit of patients drawn from new sites or acquisition batches.

## C.20 COARSE-FINE CONSISTENCY DIAGNOSTIC

Independent coarse and fine classifiers can disagree about the hierarchy even when each prediction is plausible in isolation. Table 27 reports their natural agreement on the same frozen OOF predictions as the main results. These auxiliary path metrics remain cell-weighted averages, whereas the main Macro-F1 table now balances patient contributions.

The main reporting metric was revised retrospectively without retraining, retuning, or reselecting checkpoints. The historical cell-weighted comparison family contrasted Soft-cascade with HCE and XGBoost at coarse and fine resolution, using 2,000 paired patient bootstrap resamples and Holm correction over four tests; none reached adjusted significance. Those tests do not apply to the

| Model                    | Input | Coarse CW-Macro-F1 $\uparrow$ | Fine CW-Macro-F1 $\uparrow$ |
|--------------------------|-------|-------------------------------|-----------------------------|
| Soft-cascade Transformer | APT   | 0.326 [0.300, 0.350]          | 0.152 [0.129, 0.166]        |
| Flat Transformer         | APT   | 0.323 [0.299, 0.344]          | 0.147 [0.120, 0.161]        |
| HCE Transformer          | APT   | 0.333 [0.310, 0.354]          | 0.151 [0.127, 0.166]        |
| XGBoost                  | APT   | 0.322 [0.298, 0.343]          | 0.143 [0.127, 0.158]        |
| Linear SVM               | APT   | 0.311 [0.296, 0.325]          | 0.111 [0.093, 0.126]        |
| Logistic Regression      | APT   | 0.299 [0.283, 0.315]          | 0.132 [0.113, 0.146]        |
| Random Forest            | APT   | 0.250 [0.228, 0.271]          | 0.092 [0.082, 0.100]        |
| Reference Correlation    | APT   | 0.172 [0.149, 0.189]          | 0.072 [0.055, 0.082]        |

Table 26: Historical cell-weighted pooled Macro-F1, with pointwise 95% patient-bootstrap intervals, for the frozen APT-only hierarchy predictions. The same predictions are reweighted in Table 1; the estimands differ. Classical auxiliary references are retained here rather than omitted from the benchmark.

subject-balanced results. Flat and auxiliary classical comparisons remain descriptive. The saved re-evaluation script reproduces the old point estimates before generating the new table, and records source-artifact hashes.

| Model                      | Fine Acc. | Root-excluded Hier. F1 | Consistency | Exact Path | Subtype Tree Dist. |
|----------------------------|-----------|------------------------|-------------|------------|--------------------|
| Soft-cascade Transformer   | 0.305     | 0.404                  | 0.812       | 0.253      | 2.431              |
| Flat FT-style Transformer  | 0.315     | 0.411                  | 0.795       | 0.257      | 2.407              |
| FT-style Transformer + HCE | 0.306     | 0.405                  | 0.834       | 0.259      | 2.418              |
| XGBoost                    | 0.323     | 0.420                  | 0.736       | 0.248      | 2.375              |

Table 27: Cell-weighted auxiliary hierarchy diagnostics on the same frozen OOF predictions as Table 1. Root-excluded hierarchical F1 is the example-averaged F1 between true and predicted {lineage, subtype} sets; the always-correct root is omitted. Exact Path requires both independent predictions to be correct. Consistency compares the mapped fine parent with the independent coarse prediction. Subtype Tree Distance uses only the predicted subtype and its mapped parent: 0/2/4 for correct/sibling/cross-lineage predictions (lower is better). *Unknown* is excluded; no constrained decoding is used.

#### C.21 ARCHIVED FEATURE-REDUNDANCY DIAGNOSTIC

We retain this audit only to document how easily the unresolved within-cohort signal can be compressed. Development-selected ten-feature panels reproduced the observed full-panel result on the same 40 patients, but random panels also performed strongly. This is neither biomarker discovery nor evidence that a reduced assay or low-cell clinical workflow is valid. Feature overlap across folds cannot distinguish a reproducible biological signature from a shared acquisition signature.

The complete historical protocol, curve, and random-panel, TreeSHAP, and cell-subsampling tables are preserved in the repository at `archive/compact_panel_audit.tex`; they are omitted from this PDF to avoid giving a secondary diagnostic the appearance of an assay-efficiency contribution. The archive records the cell-weighted LR ranker, conditional inner budget selection, foldwise refitting, and random-panel construction. Feature restriction involved refitting; inference-cell caps instead sampled cached held-out predictions after training on all development cells. Neither feature stability nor a 100-cell aggregation result establishes molecular specificity, training-data efficiency, or clinical validity. Such claims would require acquisition metadata and independent APT validation with a frozen pipeline.

#### C.22 BALANCED SAMPLING ABLATION

A development-only class-balanced-sampling variant did not outperform natural sampling after matching optimization steps and was not retained.

### C.23 EXPLORATORY LEAVE-ONE-DISEASE-OUT STRESS TEST

We retain only a leave-one-disease-out (LODO) representation stress test. For each run, all patients from one disease category are withheld from training and used only for evaluation. Table 28 shows that generalization weakens substantially relative to the standard patient-disjoint folds, especially for fine subtypes. Because there are only six disease groups and no external cohort, LODO is descriptive rather than an estimate of deployment robustness.

| Held-out Disease     | Fine Macro-F1 | Fine Acc.    | Coarse Macro-F1 | Coarse Acc.  | Tail Macro-F1 |
|----------------------|---------------|--------------|-----------------|--------------|---------------|
| BTC                  | 0.086         | 0.317        | 0.310           | 0.479        | 0.073         |
| CRC                  | 0.105         | 0.277        | 0.309           | 0.520        | 0.112         |
| GC                   | 0.063         | 0.269        | 0.281           | 0.488        | 0.057         |
| HCC                  | 0.063         | 0.246        | 0.264           | 0.383        | 0.072         |
| PC                   | 0.057         | 0.253        | 0.221           | 0.482        | 0.037         |
| <b>Macro-average</b> | <b>0.075</b>  | <b>0.272</b> | <b>0.277</b>    | <b>0.470</b> | <b>0.070</b>  |

Table 28: Leave-one-disease-out generalization for Soft-cascade Transformer over the five malignant held-out diseases. Each row withholds all patients of the named disease from training. The in-distribution reference using the same model under the standard patient-disjoint 5-fold protocol is fine Macro-F1 0.152 and coarse Macro-F1 0.326. A separate Normal-held-out run produces near-chance fine performance. LODO is reported only as an exploratory appendix stress test, not as proof that the model already solves cross-disease generalization.

We omit unmatched detector rankings. Cross-disease generalization remains unresolved and requires external cohorts with prespecified distribution shifts and patient-clustered uncertainty.